AdaptCloud: Information-Driven Adaptive Multimodal Fusion for Cloud-Robust Land Cover Classification

Huiting Li, *Student Member, IEEE*, Jingru Zhu, Tianyi Xu, Hao Wu, Jun Zhang, Jie Chen, and Guoqing Zhou

***Abstract*—The land cover classification (LCC) is essential for resource management and socioeconomic analysis but is often hindered by clouds that obscure spectra and disrupt spatial–temporal consistency. Current optical and synthetic aperture radar (SAR) fusion methods aimed at mitigating these impacts lack cloud conditioned reliability modeling, which restricts their generalization ability. To address this issue, we propose AdaptCloud, an information-driven adaptive multimodal fusion framework designed for cloud robust LCC. AdaptCloud mitigates cloud-induced information degradation by constructing an information exploitation chain that treats cloud cover as a limited information condition, progressively enhancing the class discriminability, observational completeness, and overall reliability. Specifically, for enhanced category discrimination, AdaptCloud leverages prior information from foundational model to strengthen feature extraction capabilities under cloud occlusion, while category adaptive fine-tuning optimizes feature extraction for specific category. By leveraging occlusion stable low-frequency optical structures and high-frequency SAR texture priors, AdaptCloud performs bidirectional injection and pixel-level multiscale fusion to improve information integrity. Furthermore, to enhance the key information reliability, AdaptCloud aligns modalities via cross-modality attention and integrates sparse filtering to preserve stable features and suppress degradation, enabling robust multimodal fusion under diverse cloud conditions. Systematic evaluations on four cloud occluded datasets, DDHRNet Korea, DDHRNet Xi'an, PIE-RGB-SAR, and CSU-OPT-SAR, show that AdaptCloud outperforms state-of-the-art (SOTA) baselines for land-cover mapping under cloud cover, achieving mean intersection over union (mIoU) gains of 0.71%–19.64% and demonstrating robust performance in cloud prone regions. In additio, the CSU-OPT-SAR dataset is established to overcome the lack of multiseasonal data, demonstrating AdaptCloud's strong generalization for cross-regional and cross-temporal LCC.**

***Index Terms*—Adaptive multimodal fusion, bidirectional information injection, cloud robust, information–exploitation chain, land cover classification (LCC).**

I. Introduction

ACCURATE land cover classification (LCC) maps are in high demand, as they are essential for understanding resource distribution and socio-economic impacts [1], [2]. With the advancement of remote sensing technology and the increasingly extensive spatial coverage of satellite imagery, remote sensing has become the dominant means for producing large-scale land cover product (LCP) [3], [4]. Representative examples include the 10-m global LCP derived from Sentinel-2 [5], the 30-m global LCP dynamics dataset spanning 1985–2022 based on the Landsat imagery [6], SinoLC-1 integrates multitemporal Google Earth imagery to generate 1-m LCP for China [6], and the 1-m LCP for six U.S. states derived from aerial images using a U-Net Transformer model [7]. Nevertheless, the acquisition of cloud-free imagery remains a major challenge, particularly in humid and high-rainfall regions. To address this, most LCP rely on multitemporal or seasonal imagery [8], [9], which allows the generation of cloud-free mosaics but inevitably compromises temporal accuracy and reduces product reliability [10], [11]. Therefore, developing high-accuracy LCC methods capable of overcoming cloud obscuration is urgently needed to break the temporal bottleneck of existing products and to improve their timeliness, accuracy, and application value.

To mitigate the impact of cloud occlusion on LCC, existing research has proposed numerous approaches. Widely used strategies include joint cloud removal and knowledge distillation from cloud-free imagery to improve LCC in cloud-covered scenarios [12], [13]. These methods can improve the classification accuracy, but the effectiveness of cloud removal when transferring between different regions remains uncertain [14], [15]. To address these limitations, time-series-based strategies have been explored, such as combining spectral indices with deep models and temporal reconstruction [16],

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Data is available on-line at https://github.com/orgs/ChenjieCSU/repositories
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[17] or applying random forests and recurrent neural network (RNN) networks to fill cloud-covered pixels from other dates [18], [19]. These approaches improve the mapping accuracy in cloudy regions; yet, their reliance on dense historical archives and frequent observations constrains large-scale and near-real-time applications. An alternative line of research directly employs cloudy imagery to preserve temporal fidelity. For example, GLNet [20] embodies this approach by leveraging discriminative features under both clear and cloudy conditions and by imposing interclass dispersion and intraclass compactness constraints. Shrestha et al. [21] used topographic parameters to mitigate the impact of cloud cover on LCC in tropical regions. Direct LCC in cloud-occluded scenes shows potential, but its stability and transferability are curtailed by sustained dependence on optical modalities. Consequently, researchers are exploring the integration of multisource data beyond optical sources to supplement and enhance LCC robustness.

Given that synthetic aperture radar (SAR) data exhibit strong resistance to cloud and fog interference while covering extensive areas, the combined classification of optical and SAR data provides an effective approach to counteract the impact of cloud cover [22], [23]. Early fusion methods mainly are simple feature concatenation that often introduced irrelevant information due to the significant cross-modal heterogeneity between optical and SAR data, resulting in the suboptimal LCC performance [24], [25]. Recent multimodal classification techniques, such as SA-Gate [26], MCANet [27], and asymmetric semantic alignment network (ASANet) [28], have advanced integration by incorporating attention mechanisms and asymmetric semantic alignment to better align cross-modal features. Furthermore, to strengthen multilevel integration, hybrid fusion strategies that combine feature-level and decision-level fusion have been developed, further enhancing the model diversity and robustness [29], [30]. Despite these advances, these methods have been introduced under cloud-free conditions, failing to adequately consider cloud cover-induced challenges and utilization of stable information under cloud cover, resulting in significant degradation of method LCC performance [12], [28].

As shown in Fig. 1, under cloud-covered conditions, optical imagery often retains relatively stable low-frequency macroscopic information, such as overall color and spatial distribution, from which structural prior cues can still be derived, while SAR imagery continues to provide relatively stable geometric and boundary information that supports to obtain high-frequency texture prior cues. In addition, as shown in Fig. 1(a), cloud cover exhibits highly diverse patterns; yet, most existing methods fail to consider the variability of optical information under cloud-obscured conditions. Given this background, the stark contrast between the inconsistence of optical observations [see Fig. 1(b)] and the relative stability of SAR observations results in residual information being underutilized, further enlarging cross-modal discrepancies.

From the perspective of limited information mining, we propose AdaptCloud, an information-driven adaptive multimodal fusion method designed to enhance the robustness of LCC in cloud-covered environments. Unlike the traditional

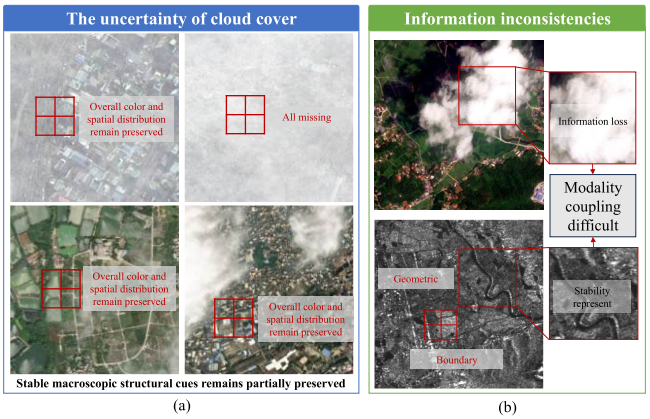


Fig. 1. Optical information available under cloud cover and the challenges of multimodal fusion. (a) Despite uncertainties in cloud cover, some low-frequency macrostructural prior cue, such as overall color and spatial distribution of features, remains exploitable. (b) Discontinuity of optical imagery and its inherent heterogeneity with SAR data further complicate the fusion process.

approach that simplistically views cloud cover as information loss, AdaptCloud defines it as a state, where unstable optical information coexists with stable SAR information and fully extracts residual information through a structured processing chain. Specifically, AdaptCloud maximizes the use of limited information and enhances class distinctiveness by leveraging foundational model prior information and category-adaptive fine-tuning. In addition, AdaptCloud supplements degraded observational data by integrating optical low-frequency structures prior to the cue with SAR high-frequency texture prior to the cue, and strengthens the selective use of reliable information under cloud uncertainty through reliability-aware feature selection. This ensures robust and efficient LCC even in cloud-covered environments. AdaptCloud outperformed seven existing state-of-the-art (SOTA) methods across four datasets covering diverse cloud conditions and demonstrated lower model performance degradation under cloudy conditions, effectively reducing reliance on cloud-free imagery. To mitigate the scarcity of multiseason, multicloud optical, and SAR datasets, this article constructed the CSU-OPT-SAR dataset using spring and summer imagery from Hunan research area, encompassing both cloud-free and cloud-contaminated conditions. The specific contributions are as follows.

1) To enhance the limited information extraction and class separability under cloud occlusion, we propose a foundation prior information-driven category disambiguation approach that leverages foundation model priors and class associations for targeted optimization, sharpening class boundaries, and improving feature extraction to learn more discriminative representations, even in cloud-contaminated environments.
2) To exploit intermodal stability priors for information supplementation under cloud occlusion, we develop a structure–texture information-guided adaptive coupling strategy that leverages complementary optical low-frequency structure and SAR high-frequency texture prior cues. Through multiscale bidirectional interactions

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and pixelwise reliability modeling, it adaptively aligns and reweights modalities to recover cloud-occluded content, suppress inconsistencies, and enhance observational completeness and fusion robustness.

3) To strengthen the selective utilization of reliable information under cloud-induced uncertainty, we propose an information stability-aware feature optimization (ISFO) strategy that integrates cross-modality attention with sparse filtering to identify and preserve cloud consistent, trustworthy signals while suppressing corrupted ones, thereby ensuring robust feature learning and enhancing multimodal fusion reliability across diverse and complex cloud conditions.
4) The remainder of this article is organized as follows. Section II reviews related work on LCC. Section III presents the proposed AdaptCloud method in detail. Section IV describes the datasets used and provides a comprehensive analysis of the experimental results. Section V discusses the computational cost, parameter sensitivity, adaptability, and comparison with existing products. Finally, Section VI concludes this article with key findings and future directions.

II. Related Works

A. Land Cover Classification

LCC profoundly regulates climate and drives ecosystem evolution, while the rapid urbanization has intensified LCC transformations, resulting in biodiversity loss, food security crises, and public health risks [31], [32]. Therefore, precise monitoring of LCC is crucial for understanding the underlying mechanisms and impacts of these changes [33], [34]. In recent years, deep learning has further advanced this field through its powerful feature representation capabilities [35], [36] that include models, such as convolutional neural networks (CNNs) [37], [38], graph convolutional networks (GCNs) [39], [40], Transformers [41], [42], and Visual Transformers (ViTs) [43], [44], all of which have achieved remarkable results in LCC. Despite this, global and national-scale LCP [11], [45] often rely on the integration of multitemporal historical imagery to generate cloud-free observations, which inevitably compromises temporal accuracy and limits their capacity for real-time updates. For example, LCMAP [8] uses annual and seasonal composites, typically from spring and autumn, to generate 30-m LCP since 1985, while NLCD [46] updates every five years using seasonally composited Landsat imagery. SinoLC-1, the first 1-m national-scale land cover map of China [11], integrates multitemporal, cloud-free Google Earth imagery to achieve high-spatial completeness. Although this approach ensures seamless spatial coverage, the temporal heterogeneity of the source imagery can limit its ability to capture rapid land cover changes [47], [48]. Therefore, the development of cloud-resistant LCC methods is crucial for enhancing the temporal resolution of large-scale land cover mapping.

B. LCC Under Cloud Obscure

In recent years, addressing the challenges of cloud cover has become a central concern for achieving high-accuracy LCC. One approach involves incorporating cloud removal techniques and cloud-free image knowledge distillation into the classification process to reduce cloud interference. For instance, Xu et al. [12] proposed CloudSeg, which combines knowledge distillation with declouding methods to enhance classification accuracy. Wang et al. [13] proposed a multistep transfer strategy to adapt a source model, pretrained on cloudless optical satellite images, to a multimodal target domain with clouded optical and SAR images, enhancing cross-domain knowledge transfer. Although these methods can effectively reduce cloud-induced noise, their performance often declines under highly variable spatiotemporal conditions [49], [50]. An alternative strategy leverages time-series reconstruction to recover information lost due to cloud occlusion. Man et al. [51] synthesized dense Landsat 8 time series to generate clear images for training multiple supervised classifiers, and Li et al. [16] developed an integrated time-series mapping framework combining spectral index fusion with deep learning models. Such methods can improve the classification accuracy but generally require high-frequency observations, which are difficult to obtain in many real-world scenarios. Direct classification approaches provide another solution by automatically detecting cloud-covered pixels and extracting discriminative features, thus enhancing temporal resolution. Xie et al. [52] introduced a multitemporal SVM-based classification method that merges results through majority voting, expectation-maximization, and SVM probability to handle missing data, while Shrestha et al. [21] applied modified classifiers, including maximum likelihood, nearest neighbor, and Parzen classifiers, to account for pixels obscured by clouds or snow. Despite improving temporal accuracy, these methods often struggle to extract effective features in environments with severe or heterogeneous cloud cover.

C. Optical and SAR Fusion for LCC

Given that SAR can penetrate clouds, the integration of SAR and optical data offers a powerful solution for overcoming data gaps caused by cloud cover, significantly improving the temporal consistency and LCC timeliness. Optical and SAR fusion LCC mainly uses three paradigms, pixel-level fusion, feature-level fusion [53], and decision-level fusion [54]. Pixel-level fusion involves directly combining pixel information from both optical and SAR data sources, but aligning these heterogeneous modalities presents significant challenges [55], [56]. More recently, feature-level fusion has gained traction as it involves extracting features from both modalities and combining them into a unified feature set, improving the robustness of LCC [57], [58], especially in regions affected by cloud cover. For example, Zhang et al. [59] introduced a multilevel fusion of optical and SAR data within a multiscale feature extraction framework, developing an SVM model that significantly enhanced LCC accuracy and robustness in high-cloud regions; Zhang et al. [28] proposed a novel ASANet architecture that leverages the complementary features of SAR and optical data, achieving strong performance even in cloud-covered scenarios. Decision-level fusion aims to integrate the outputs from different classifiers and feed them into another classifier to obtain the final prediction result [60]. For instance,

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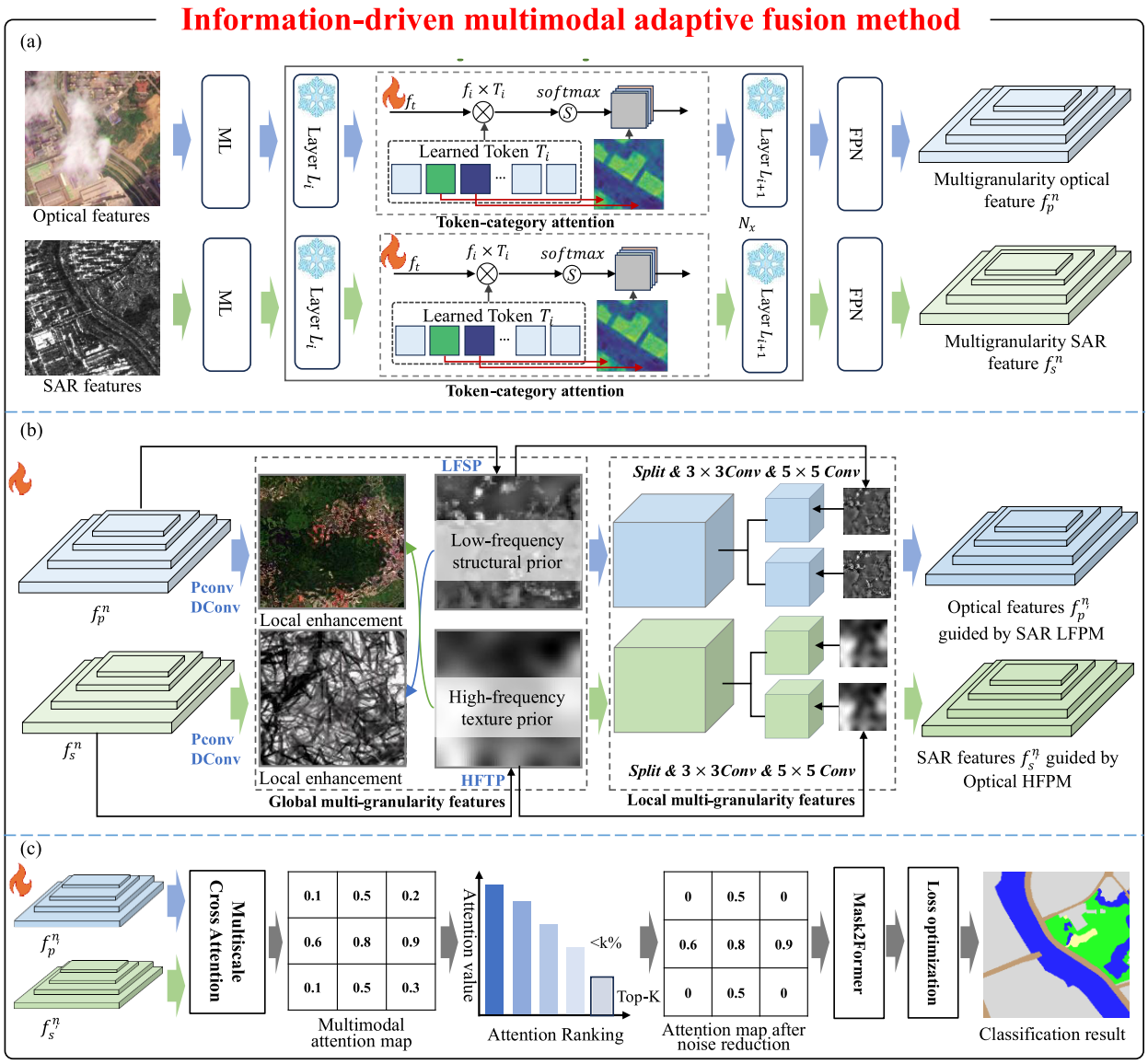


Fig. 2. Overview of the proposed AdaptCloud method. FICD is designed to enhance feature extraction capabilities and class discrimination capabilities; SIAC aims to improve observational completeness and fusion robustness; ISFO enhances the selective exploitation of reliable information under cloud-induced uncertainty; HFTP is to obtain a high-frequency texture prior cue of SAR feature; LFSP is to obtain a low-frequency structure prior cue of optical feature. (a) Foundation-prior information-driven category disambiguation. (b) Structure texture information guided adaptive coupling. (c) Information stability-aware feature optimization.

Bigdeli and Pahlavani [61] proposed a decision-level fusion framework based on the synergy of multisource remote sensing data to achieve the fusion and optimization of heterogeneous modal features at the decision level. Liu et al. [29] proposed SoftFormer, integrating CNN-based local feature extraction with Transformer-based global context modeling for multilevel feature fusion and concurrent decision- and feature-level integration. Despite some progress, most of these methods were proposed under cloudless conditions and did not fully consider the challenges posed by cloud cover or the utilization of stable information beneath the cloud layer, resulting in a significant decline in the performance of LCC methods. Although fusing SAR images can significantly enhance the timeliness of LCP, existing fusion methods primarily rely on cloud-free images and seldom account for the information loss caused by cloud cover, resulting in challenges in multimodal image fusion under cloudy conditions.

III. Proposed Method

The overall framework of AdaptCloud (see Fig. 2) adopts a dual-branch encoder–decoder architecture, where optical and SAR features are independently encoded and then fused. From the perspective of limited information mining, the framework incorporates three tailored modules to progressively exploit residual cues from category ambiguity, observational integrity, and information reliability. First, foundation-prior information-driven category disambiguation (FICD) builds upon a ViT encoder initialized with CLIP pretrained weights, where low-rank adaptation (LoRA)-based fine-tuning on category-associated tokens constructs a category-specific attention

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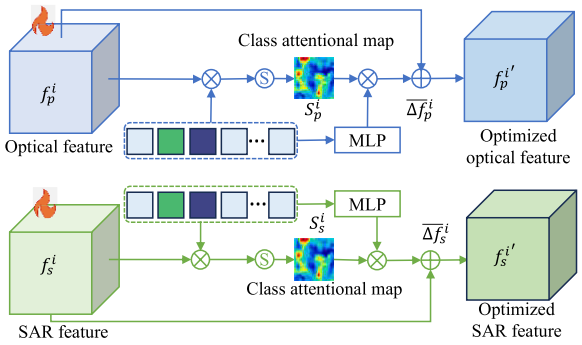


Fig. 3. FICD module. Blue indicates optical features, while green indicates SAR features. The differently colored squares within the blue box represent optical learnable tokens; the differently colored squares within the green box represent SAR learnable tokens.

map to enhance the category discriminability. Second, the structure–texture information-guided adaptive coupling (SIAC) applies the complementarity between optical low-frequency structural and SAR high-frequency texture prior cues through multiscale bidirectional interactions. This design mines incomplete observational information and compensates for missing information in cloud-covered regions. Third, the ISFO aligns and enhances key spatial–spectral features through cross-modal attention, subsequently ranks Top-K features based on attention maps to suppress cloud occlusion uncertainty and SAR noise interference, thus extracting stable and reliable multimodal complementary representations under cloud-obscured conditions. Finally, Mask2Former was selected as the segmentation head due to its robust global modeling capabilities and ability to capture multiscale information [62], which are crucial for addressing spatial discontinuities and object confusion under cloud-covered conditions.

A. Foundation-Prior Information-Driven Category Disambiguation

To fully exploit limited information under cloud obscuration and enhance class differentiation, we propose FICD, which inherits priors from the foundation model to strengthen the feature representation and employs trainable tokens to construct class-specific attention maps for optimizing category discrimination. Specifically, inspired by Rein approach [63], as shown in Fig. 3 we freezes the parameters of all layers and only fine-tunes the trainable token sequence, allowing the model to quickly focus on residual information under cloud cover conditions. In addition, we optimize each modality using independent encoders that align the optical and SAR data toward a common semantic category, ensuring that both modalities share consistent semantic representations despite cloud interference. First, learnable tokens are introduced for both optical and SAR modalities. Subsequently, the optically trainable token $T_p^i = \{T_p^i \in \mathbb{R}^{m\times c} | i \in \mathbb{N}, 1 \le i \le N\}$ perform elementwise multiplication with the original optical features, quantitatively evaluating the association between different tokens and feature vectors. Through alignment with unique category instances via the Softmax function, thus generating category attention map S_p^i. Similarly, the SAR trainable token $T_s^i = \{T_s^i \in \mathbb{R}^{m\times c} | i \in \mathbb{N}, 1 \le i \le N\}$ is multiplied by the original SAR features, producing the SAR category attention map S_s^i through the Softmax function

$$S_p^i = \text{Softmax}\left(\frac{f_p^i \times \left(T_i^p\right)^T}{\sqrt{c}}\right) \tag{1}$$

$$S_s^i = \text{Softmax}\left(\frac{f_s^i \times \left(T_i^s\right)^T}{\sqrt{c}}\right) \tag{2}$$

where T_p^i represents the sequence of optical tokens in layer i. T_s^i represents the sequence of SAR tokens in layer i. S_p^i is the attention map of the optical feature. S_s^i is the attention map of the SAR feature.

Due to the softmax function, the sum of S_p^i or S_s^i equals 1; however, when all features are precise, this may lead to unnecessary variation. To address this issue, $S_p^i(:, 2:m)$ and $S_s^i(:, 2:m)$, respectively, select columns 2–m from S_p^i and S_s^i, while $T_p^i(2:m)$ and $T_s^i(2:m)$, respectively, select rows 2–m from T_p^i and T_s^i. This strategy prevents unnecessary adjustments by initially assigning a higher value to the first token and discarding it afterward. This ensures that the sum of S_p^i and S_s^i for each row remains between 0 and 1, thereby minimizing the risk of inappropriate modifications. Through the above methods, the attention maps S_p^i and S_s^i are used to weight the original token features processed by the multilayer perceptron (MLP) layer, generating the category-associated incremental representations $\Delta\overline{f_p^i}$ and $\Delta\overline{f_s^i}$, which are then added to the i layer original features f_p^i and f_s^i obtaining optimized optical feature f_p^i and SAR feature f_s^i

$$\Delta\overline{f_p^i} = S_p^i(:, 2:m) \times \left[T_p^i(2:m) \times W_{T_p^i} + b_{T_p^i}\right] \tag{3}$$

$$\Delta\overline{f_s^i} = S_s^i(:, 2:m) \times \left[T_s^i(2:m) \times W_{T_s^i} + b_{T_s^i}\right] \tag{4}$$

$$f_p^{i'} = f_p^i + \Delta\overline{f_p^i} \tag{5}$$

$$f_s^{i'} = f_s^i + \Delta\overline{f_s^i} \tag{6}$$

where W_{T_i} and b_{T_i} denote the weight and bias of the MLP, respectively. $S_s^i(:, 2:m)$ is designed to choose columns 2–m of SAR attention map S_s^i, and $T_s^i(2:m)$ denotes the selection of rows 2–m of the optical trainable token T_s^i; $\Delta\overline{f_p^i}$ is an optical feature increment; $\Delta\overline{f_s^i}$ is the increment of the SAR feature; $f_p^{i'}$ is an optimized optical feature; $f_s^{i'}$ is optimized SAR feature.

B. Structure–Texture Information-Guided Adaptive Coupling

To exploit intermodal stable prior information for mitigating information loss caused by cloud occlusion, the SIAC model is proposed. By leveraging the stable and complementary structural and texture prior cues between optical and SAR data, SIAC establishes a cross-modal alignment and coupling mechanism to improve the completeness of missing information. While cloud cover obscures fine details, optical imagery generally retains valuable low-frequency information, such as overall color and spatial distribution, even in partially occluded regions. The SIAC model utilizes this optical low-frequency structural prior cue to guide SAR feature extraction, thereby compensating for the limited low-frequency spectral content in SAR data arising from its imaging mechanism.

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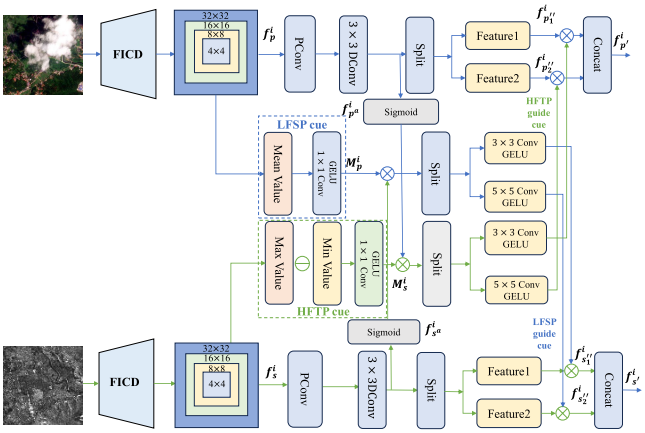


Fig. 4. SIAC module. This section inputs optical and SAR features from four scales that have been fine-tuned using FICD. PConv represents pointwise convolution. DConv represents the depthwise convolution.

Meanwhile, the SAR high-frequency texture prior cue is used to guide the recovery of optical features, particularly textures and local details that are more susceptible to cloud interference. This dual prior-guided mechanism facilitates multimodal fusion, yielding more complete and robust feature representations under cloud-affected conditions. As shown in Fig. 4, SIAC consists of two parallel branches and two prior cross branches. The global mean aggregates spatial responses over the entire feature map, thereby emphasizing large-scale, spatially coherent patterns associated with overall color and spatial distribution and is therefore employed as the guidance information for the optical low-frequency structural prior cue. In contrast, the max–min residual characterizes the amplitude range and contrast of feature responses, making it more sensitive to local variations, such as geometric and edges and is thus adopted as the guidance information for the SAR high-frequency texture prior cue. Both prior cues are further enhanced by Gaussian error linear unit (GELU) to generate more stable and discriminative prior attention maps for subsequent fusion. Consequently, the optical low-frequency structural prior cue improves structural consistency and overall robustness, while the SAR high-frequency texture prior cue enhances local details and contrast, with the two jointly producing a synergistic effect

$$\underset{n=1,\ldots,4}{M_p^n} = \mathrm{GELU}\left(\mathrm{mean}\left(\mathrm{FICD}\left(f_p\right)\right)\right) \tag{7}$$

$$\underset{n=1,\ldots,4}{M_s^n} = \mathrm{GELU}\left(\max\left(\mathrm{FICD}\left(f_s\right)\right) - \min\left(\mathrm{FICD}\left(f_s\right)\right)\right) \tag{8}$$

where f_p is the optical feature and f_s is the SAR feature; n represents different scales and GELU is the Gaussian error linear unit. M_p^n is the low-frequency structure prior cue of the optical feature, and M_s^n is the high-frequency texture prior cue of the SAR feature.

In the optical branch, PConv is first used to expand the receptive field and channel dimensions, adaptively capturing large-scale contextual information across multiple spatial scales. This is particularly effective in cloud-occluded environments, compensating for image gaps caused by cloud cover and restoring key background details. DConv is then applied to refine local details, accurately reconstructing fine-grained features in occluded regions, enhancing object recognition. The process is optimized by multiplying the attention map $f_{p^a}^n$, generated via an activation function, with the DConv-processed SAR high-frequency texture prior cue M_s^n, yielding globally enhanced high-frequency texture prior information and improving the recovery of optical features, especially under cloud occlusion. To further improve the robustness in complex cloud-occluded environments, SAR high-frequency texture prior cue is split into two levels $M_{s_1'}^n$ and $M_{s_2'}^n$, extracting global and local high-frequency texture prior cue. Lower scale convolutional kernels capture broad details for large-area recovery, while higher scale kernels focus on fine-grained features, effectively restoring small-scale details occluded by clouds. Finally, the optical features $f_{p'}^n$, extracted under SAR residual a priori guidance, are obtained through a Split operation

$$f_{p^a}^n = \underset{3\times3}{\mathrm{DConv}}\left(\mathrm{PConv}\left(f_p^n\right)\right) \tag{9}$$

$$f_{p_1'}^n = \mathrm{Split}\left(f_{p^a}^m\right); f_{p_2'}^m = \mathrm{Split}\left(f_{p^a}^m\right) \tag{10}$$

$$f_{p^a}^n = \mathrm{Sigmoid}\left(\mathrm{DConv}\left(\mathrm{PConv}\left(f_p^n\right)\right)\right) \tag{11}$$

$$M_{s_1'}^n = \underset{3\times3}{\mathrm{Conv}}\left(\mathrm{Split}\left(\mathrm{PConv}\left(M_s^n\right) * f_{p^a}^n\right)\right) \tag{12}$$

$$M_{s_2'}^n = \underset{3\times3}{\mathrm{Conv}}\left(\mathrm{Split}\left(\mathrm{PConv}\left(M_s^n\right) * f_{p^a}^n\right)\right) \tag{13}$$

$$f_{p_1''}^n = f_{p_1'}^n * M_{s_1'}^n; f_{p_2''}^n = f_{p_1'}^n * M_{s_1'}^n \tag{14}$$

$$f_{p'}^n = \mathrm{Concat}\left(f_{p_1''}^n, f_{p_2''}^n\right) \tag{15}$$

where f_p^n is the optical feature after FICD, $f_{p^a}^n$ is the optical feature after the activation function, M_s^n is the high frequency priori cue of SAR, $M_{s_1'}^n$ and $M_{s_2'}^n$ are the attention map obtained a priori cue from optical features and SAR feature, $f_{p_1''}^n$ and $f_{p_1''}^n$ stand for SAR high-frequency prior-guided optical features, and $f_{p'}^n$ is the feature after splicing.

In the SAR branch, PConv is first used to expand the channel dimension of the features, enabling the model to adaptively capture structural information at different scales. Next, DConv is used to further refine local details and accurately recover features in cloud-obscured areas. The processed SAR features generate an attention map $f_{s^a}^n$, which is then multiplied with the optical low-frequency structure prior features $M_{p_1'}^n$ via an activation function to incorporate this information into the SAR feature optimization process. This process enables the SAR features to better integrate the background information provided by the optical image. The optical low-frequency structure prior is then split into two levels $M_{p_1'}^n$ and $M_{p_2'}^n$, and convolutional kernels of different scales are used to extract global and local high-frequency texture prior information, respectively. The lower scale convolutional kernel captures broader information, helping to recover large-scale details under cloud obscuration, while the higher scale convolutional kernel focuses on feature extraction at the detailed level, effectively recovering the fine information in smaller areas. Finally, SAR Features 1 and 2, guided by the global and local priors, are concatenated to form the enhanced SAR features $f_{p'}^n$. These enhanced features effectively improve the performance of SAR images in cloud-occluded environments and optimize the accuracy of ground object recognition, especially in

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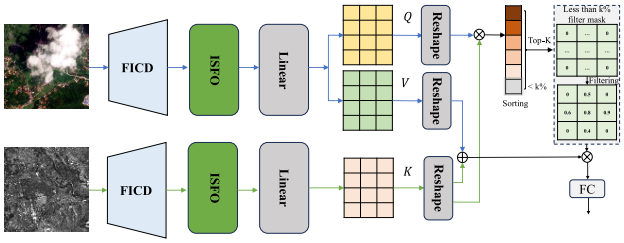


Fig. 5. ISFO module. This section inputs four scale-specific features processed via SSIM. Subsequently, a cross-attention mechanism focuses on important spatial features across different scales. Sparsity enhancement selects the Top-K% key features to eliminate noise, ultimately retaining only stable features.

cloud-occluded scenes, ensuring the robustness and accuracy of the model

$$f_{s^a}^n = \underset{3\times3}{\mathrm{DConv}}\left(\mathrm{PConv}\left(f_s^n\right)\right) \tag{16}$$

$$f_{s'_1}^n = \mathrm{Split}\left(f_{p^a}^n\right); f_{s'_2}^n = \mathrm{Split}\left(f_{s^a}^n\right) \tag{17}$$

$$f_{s^a}^n = \mathrm{Sigmoid}\left(\mathrm{DConv}\left(\mathrm{PConv}\left(f_s^n\right)\right)\right) \tag{18}$$

$$M_{p'_1}^n = \mathrm{GELU}\left(\underset{3\times3}{\mathrm{Conv}}\left(\mathrm{Split}\left(\mathrm{PConv}\left(M_p^n\right) * f_{s^a}^n\right)\right)\right) \tag{19}$$

$$M_{p'_2}^n = \mathrm{GELU}\left(\underset{5\times5}{\mathrm{Conv}}\left(\mathrm{Split}\left(\mathrm{PConv}\left(M_p^n\right) * f_{s^a}^n\right)\right)\right) \tag{20}$$

$$f_{s'_1}^n = \mathrm{PConv}\left(\mathrm{DConv}\left(\mathrm{Gelu}\left(\mathrm{Split}\left(f_{s'}^n\right)\right)\right)\right) * M_{p'_1}^n \tag{21}$$

$$f_{s'_2}^n = \mathrm{PConv}\left(\mathrm{DConv}\left(\mathrm{Gelu}\left(\mathrm{Split}\left(f_{s'}^n\right)\right)\right)\right) * M_{p'_2}^n \tag{22}$$

$$f_{s'}^n = \mathrm{Concat}\left(f_{s'_1}^n, f_{s'_1}^n\right) \tag{23}$$

where f_s^n is the SAR feature after FICD, $f_{s^a}^n$ is the SAR feature after the activation function, M_p^n is the low-frequency priori information of SAR, $M_{p'_1}^n$ and $M_{p'_2}^n$ are the attention map obtained a priori from optical features versus SAR, $f_{s''_1}^n$ and $f_{s''_1}^n$ stand for SAR high-frequency prior-guided optical features, and $f_{s'}^n$ is the feature after splicing.

C. Information Stability-Aware Feature Optimization

To enhance the use of stable information to reduce the cloud cover uncertainty impacts, which causes key features to vary across conditions, we propose ISFO, integrating cross-modal attention with sparsity-inducing attention to dynamically select and refine critical information from optical and SAR modalities. The ISFO computes feature similarities, using the Top-K algorithm to select the most significant features. This approach adapts to varying cloud cover, mitigating feature displacement and improving model robustness and accuracy under dynamic cloud occlusion. Specifically, as shown in Fig. 5, the linear transformation of optical and SAR feature are performed to map the input features to the heterogeneous semantic SIAC, which generates the query (Q_p^n), key (K_s^n), and value (V_p^n) vectors. Subsequently, the similarity between the K_s^n and Q_p^n features is computed through matrix multiplication to generate a feature attention graph A^n. This feature attention map is specifically designed to capture feature dependencies within and between modalities, thereby enhancing the model's ability to represent alignment between features

$$Q_p^n = f_{p'}^n W_{Q_p^i}^T, K_s^n = f_{s'}^n W_{K_s^i}^T, V_p^n = f_{p'}^n W_{V_p^i}^T \tag{24}$$

$$A^n = \frac{Q_p^n \cdot \left(K_s^n\right)^T}{\sqrt{d_k}} \tag{25}$$

where $W_{Q_p^n}^T$, $W_{K_s^n}^T$, and $W_{V_p^n}^T$ indicate the learnable parameters, $f_{p'}^n$ is the optical features extracted by FICD and SIAC, and $f_{s'}^n$ is the SAR features extracted by FICD and SIAC.

Subsequently, attention map A^n is sorted, and undisturbed features are selected based on the k-value. The value of k is determined by multiplying the retention ratio coefficient by the length of the feature sequence. A Top-K filter mask M_f^n is then generated, with the corresponding regions set to a mask value of 0. Multiply the attention map A^n by the Top-K filter mask M_f^n to obtain the attention map A_{topk}^n after suppressing cloud cover and noise interference. Finally, to ensure the balanced integration of optical and SAR features and enhance stable complementary features, we apply $V_p^n + K_s^n$ for fusion. The fused features are then multiplied by the attention map A_{topk}^n to suppress cloud and noise interference, yielding more stable multimodal features F^n

$$K = \tau^* \mathrm{L_{ref}} \tag{26}$$

$$A^n = \{a_1, a_2, \ldots, a_d] \tag{27}$$

$$M_f^n = \mathrm{A_{sort}}\{a_1, a_2, \ldots, a_d\} \times K \tag{28}$$

$$A_{topk}^n = A^n \times M_f^n \tag{29}$$

$$F^n = \left(V_p^n + K_s^n\right) \times A_{topk}^n \tag{30}$$

where τ is retention ratio coefficient, L_{ref} is the length of the feature sequence, A^n denotes the feature map, $\{a_1, a_2, \ldots, a_d\}$ is the feature value within the feature map, $\mathrm{A_{sorted}}$ denotes the ascending order of the feature values along the specified dimension (dim), M_f^n is Top-K filter mask, and A_{topk}^n denotes the Top-K% of significant features selected.

IV. Results

A. Experimental Design and Setup

To comprehensively validate the effectiveness and generalization of AdaptCloud, the experimental design was structured around complementary perspectives reflecting both methodological verification and practical applicability. On the one hand, to assess the classification capability under cloud-covered conditions, experiments were conducted on cloud-obscured data with quantitative evaluation metrics and qualitative visualizations, benchmarked against seven SOTA multimodal semantic segmentation methods (FuseNet [30], SA-Gate [26], MFNet [64], FTransUNet [65], CMFNet [66], ASANet [28], and DCIFNet [67]) that represent prevailing architectures such as SAM, CLIP, Transformer, ConvNext, and CNN. On the other hand, the mapping performance was examined across cloud-covered using the CSU-OPT-SAR dataset, and ablation experiments were performed to verify the contribution of each individual module.

AdaptCloud is implemented using the MMSegmentation framework, employing CLIP as its pretrained foundation model. Its backbone network apply VIT, while the segmentation head adopts Mask2Former. To thoroughly evaluate the performance of the proposed method, all experiments use 20% of the samples for training and 80% for testing. All the training and testing of the models in this article were done on a Quadro RTX 4090 GPU platform under Ubuntu 18.04 Linux operating system. A total of 1 00 000 iterations were executed

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TABLE I
SETTINGS OF HYPERPARAMETERS FOR EACH METHOD

Model	Optimizer	Ir	Momentum/Betas	Weight-decay
FTransUNet [65]	SGD	0.01	0.9	0.0005
CMFNet [66]	SGD	0.01	0.9	0.0005
FuseNet [30]	SGD	0.01	0.9	0.0005
MFNet [64]	SGD	0.01	0.9	0.0005
SA-Gate [26]	SGD	0.01	0.9	0.0005
ASANet [28]	AdamW	0.0001	(0.9,0.999)	0.05
DCIFNet [67]	AdamW	0.00006	(0.9-0.999)	0.01
AdaptCloud	AdamW	0.0001	(0.9, 0.999)	0.05

TABLE II
DETAILED INFORMATION FOR EACH DATASET

Dataset	location	Climate zones	Coverage area (km^2)	Resolution (m)	Cloud cover characteristics	category
DDHRNet korea [68]	Pohang	Subtropical humid climate	94.37	1	Including thin cloud and thick cloud cover	5
DDHRNet Xi'an [68]	Xi'an	Warm temperate semi-humid continental monsoon climate	8.85	1	Mainly thick cloud cover	5
PIE-OPT-SAR [28]	Pearl River Delta	Subtropical monsoon climate	2,869.49	3	Including thin cloud and thick cloud cover	6
CSU-OPT-SAR	Changsha; Zhuzhou	Subtropical monsoon climate	4,113.22	3	Including thin cloud and thick cloud cover	5

with validation at 2000 intervals and the same training and testing data were used for all experiments. To ensure the fairness of the experimental results, all experiments in this article strictly follow the unified experimental configuration. The evaluation metrics are chosen to be mean intersection over union (mIoU), Kappa coefficient, and overall accuracy (OA), which are commonly used in semantic segmentation tasks. All model results are based on the accuracy of our own replications. Table I lists the hyperparameter settings for all methods.

B. Dataset

To systematically evaluate AdaptCloud, we used four paired optical and SAR datasets from Pohang, South Korea, and from Xi'an, the Pearl River Delta, and Hunan, China. These datasets cover diverse regional contexts, including coastal and inland areas, temperate and subtropical climates, and varying degrees of urbanization, encompassing primary land cover types, such as cropland, forest, water bodies, and impervious surface. More importantly, these datasets encompass multiple cloud conditions and their seasonal variations, enabling rigorous testing of the model's discriminative power. The specific characteristics of each region are detailed below, and the corresponding dataset information is summarized in Table II.

1) Dataset of Pohang Region in Korea: The DDHRNet Korea dataset [68] provides fused 1-m resolution GF2 optical and GF3 SAR imagery. As shown in Fig. 6, the DDHRNet-Korea subset contains optical images under multiple cloud conditions, including thin-cloud, thick-cloud, and cloud-free scenarios, along with corresponding SAR data. Its spatial extent is 10 240 × 9216 pixels, covering 94.37 km^2, with

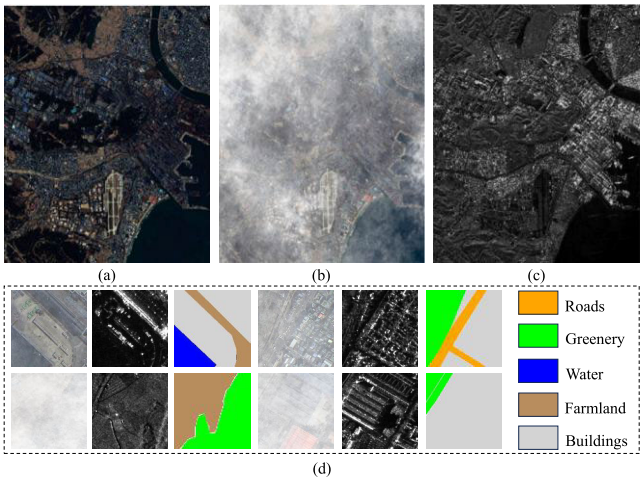


Fig. 6. Example samples from the DDHRNet-Korea dataset. (a) No cloud-covered optical image. (b) Cloud cover optical image. (c) SAR image. (d) Sample example.

land-cover proportions of 41% buildings, 12% greenery, 23% wetlands, 14% impervious surfaces, and 5% roads.

2) Dataset of Xi'an Region in China: This DDHRNet Xi'an dataset [68] includes cloud and cloud-free images. Fig. 7 presents the DDHRNet-Xi'an dataset, primarily composed of thick cloud scenes, with an image resolution of 2304 × 3840 pixels and a coverage area of 8.85 km^2. The land cover distribution includes 31% buildings, 37% greenery, 19% wetlands, 4% buildings, and 7% roads.

3) Dataset of Pearl River Delta Region in China: The data for this study are derived from the PIE-RGB-SAR dataset [28].

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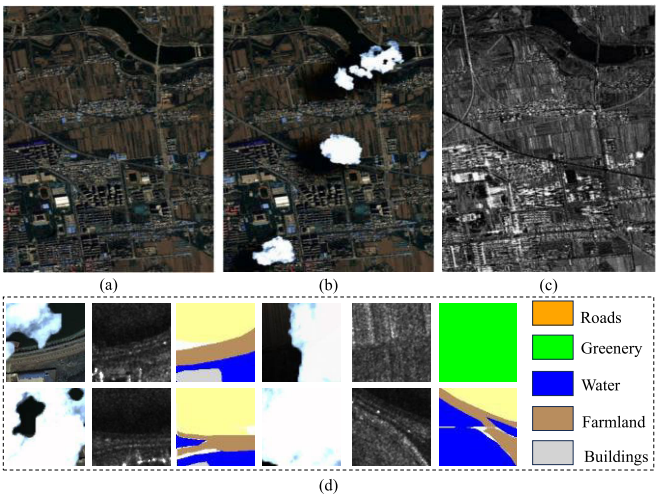


Fig. 7. Illustrative samples from the DDHRNet-Xi'an dataset. (a) No cloud-covered optical image. (b) Cloud cover optical image. (c) SAR image. (d) Sample example.

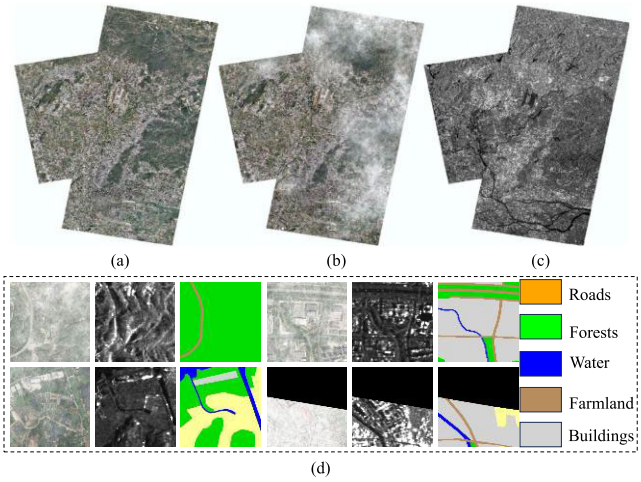


Fig. 8. Sample example of PIE-RGB-SAR dataset. (a) No cloud covered optical image. (b) Cloud cover optical image. (c) SAR image. (d) Sample example.

As illustrated in Fig. 8, the PIE-RGB-SAR dataset comprises cloudy (including both thin and thick clouds) and cloud-free optical images, along with corresponding SAR imagery, exhibiting typical characteristics of cloud-induced interference. It encompasses six land cover categories: background (7.99%), city (29.61%), roads (3.23%), water (7.26%), forests (42.13%), and cropland (9.77%). The dataset contains 4865 images, each with a resolution of 256 × 256 pixels, covering an area of 2869.49 km^2.

4) Dataset of Changsha Region in China: Through data preprocessing and calibration, we construct a paired optical-SAR dataset, termed CSU-OPT-SAR, using 3-m optical imagery derived from the Level-3B products of the PlanetScope constellation. This dataset includes both cloud-affected and cloud-free optical images alongside corresponding SAR data, captured under diverse multisource and multimeteorological conditions. The sample sizes for summer and winter are 11 630 × 16 550 pixels, covering an area of 1732.29 km^2, and 10 748 × 12 376 pixels, covering 1197.16 km^2, respectively. The data covered farmland (19.74%), green spaces (44.44%), impervious surfaces (16.04%), roads (7.51%), and water bodies (10.89%). In addition, two unlabeled image sets 12 032 × 7936 which cover 859.37 km^2 and 6400 × 5632 pixels which cover 324.40 km^2 were acquired which can be used classification testing and qualitative assessment of cloud removal tasks. As illustrated in Fig. 9, the dataset covers different acquisition times and spatial extents, demonstrating strong intertemporal and cross-regional diversity. Notably, the study area is situated in hilly regions with a subtropical monsoon climate, where cultivated land and buildings are sparsely and irregularly distributed. Seasonal variations further complicate classification by increasing confusion between cropland and green land, while frequent cloud and fog cover exacerbate the challenge of long-term, high-frequency land cover mapping. Consequently, this dataset offers valuable and robust support for research on multisource remote sensing classification and intertemporal land cover mapping under complex meteorological conditions.

C. Comparative Analysis of Results

To evaluate AdaptCloud's LCC performance under cloud cover, this section compares it with seven SOTA methods on cloud cover datasets through quantitative and qualitative analyses, including DDHRNet-Korea, DDHRNet-Xi'an, PIE-OPT-SAR, and CSU-OPT-SAR.

1) Results on DDHRNet Korea Dataset: The quantitative comparison results between AdaptCloud and competing methods on the DDHRNet Korea dataset are presented in Table III. AdaptCloud achieves the highest classification accuracy, with mIoU scores exceeding those of FTransUNet, CMFNet ASANet, FuseNet, SA-Gate, MFNet, and DCIFNet by 67.78%, 50.88%, 45.17%, 32.79%, 22.99%, 21.14%, and 10.71% in mIoU, respectively. These results reveal that existing multimodal models rely on complete optical features and fail to exploit residual information under cloud occlusion. Therefore, models such as FTransUNet, CMFNet, ASANet, and FuseNet encounter significant difficulties during convergence particularly in categories with sparse information like roads and farmland. In these scenarios, visual signals are often weak or fragmented, fully illustrating their inadequate adaptability to cloud-occlusion scenarios. In contrast, AdaptCloud achieved an OA improvement of 5.06% compared with the strongest baseline model, highlighting SIAC's capability to compensate for missing observational data and maximize the utilization of limited residual clues. Although DCIFNet progressively refines optical and SAR representations via cross-modal structural consistency to suppress cloud-induced noise, its limited exploitation of macroscopic optical context (e.g., low-frequency components and global tonal cues) leads to inferior classification performance relative to AdaptCloud under cloud occlusion. Cloud occlusion frequently causes boundary information disruption, resulting in discontinuous road segmentation and misalignment of multimodal features, which AdaptCloud mitigates by enhancing the information integrity and preserving structural continuity. Under conditions

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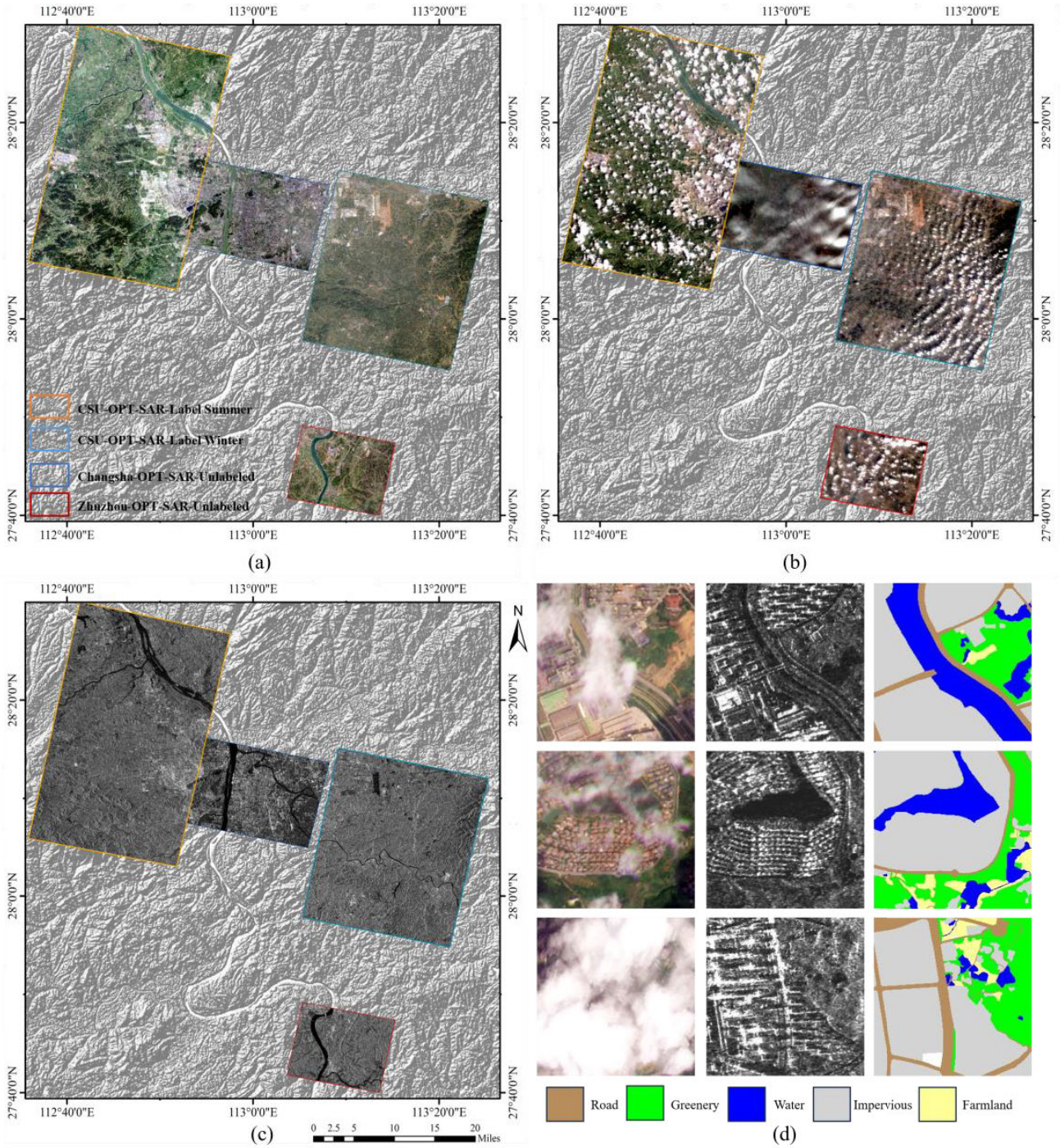


Fig. 9. CSU-OPT-SAR dataset distribution and data characteristics.

of cloud cover uncertainty, the spectral ambiguity often leads to confusion between farmland and greenery, while features exhibit significant noise. AdaptCloud effectively distinguishes farmland from greenery, achieving classification accuracies 6.43% and 21.63% higher than existing SOTA methods, respectively, validating the effectiveness of FICD and ISFO in optimizing category separation and selectively retaining reliable information.

The visual comparison of classification results from different methods on the DDHRNet Korea dataset is shown in Fig. 10. As shown in Fig. 10, AdaptCloud demonstrates superior performance in overall classification results compared with other methods under varying degrees of cloud cover, proving that our approach can effectively extract residual features from optical and SAR modalities. When color distortion and edge blurring occur in thin cloud scenes [see Fig. 10(a) and (b)], AdaptCloud integrates high-frequency SAR prior information into a multiscale bidirectional interaction mechanism that enables clearer boundary delineation and reduced errors. For thick-cloud scenarios

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TABLE III
CLASSIFICATION PERFORMANCE OF VARIOUS METHODS ON THE DDHRNET KOREA DATASET UNDER CLOUD-COVERED CONDITIONS

Methods	Buildings	Roads	Greenery	Water	Farmland	mIoU	OA	Kappa
FTransUNet [65]	75.44	0.00	32.65	12.61	19.06	17.72	45.51	17.72
CMFNet [66]	88.83	0.15	85.81	52.96	1.54	34.62	70.84	57.34
ASANet [28]	94.06	0.0	77.57	95.95	0.0	40.33	74.03	61.44
FuseNet [30]	89.13	5.05	70.6	95.75	62.31	52.71	76.33	65.01
SA-Gate [26]	79.90	56.36	85.73	92.25	69.30	62.51	80.55	72.83
MFNet [64]	93.49	12.79	80.90	96.05	81.95	64.36	85.31	78.75
DCIFNet [67]	93.24	53.03	87.68	98.30	81.21	74.79	89.10	84.45
AdaptCloud	95.89	81.17	92.35	99.14	92.14	**85.50**	**94.16**	**91.74**

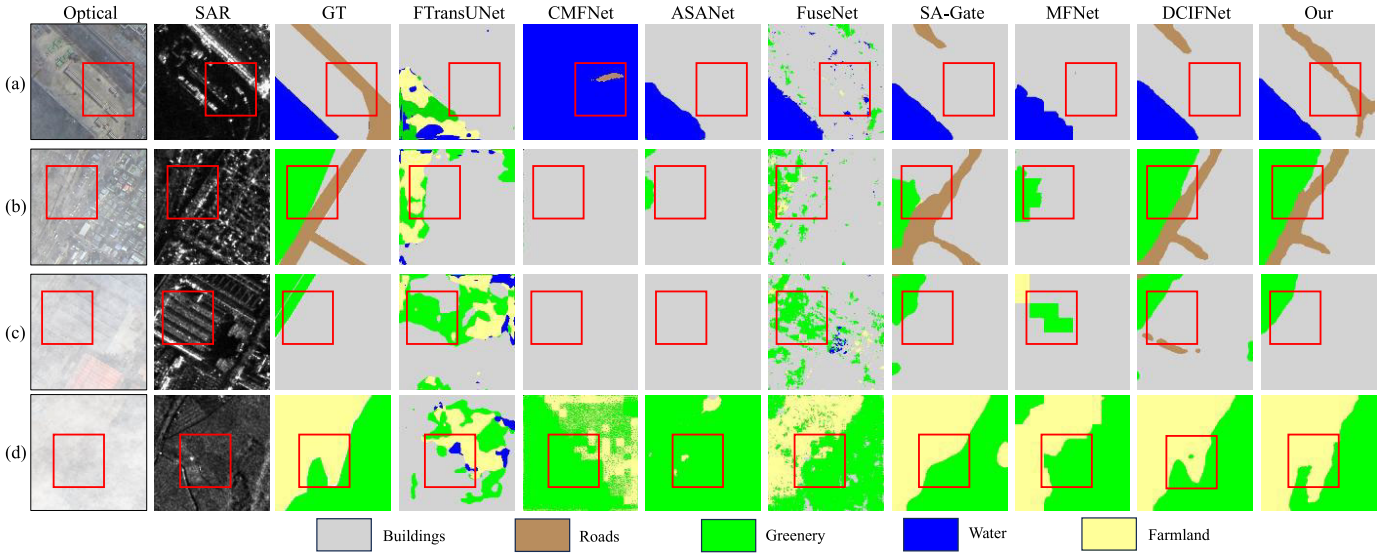


Fig. 10. Visualized classification results of DDHRNet Korea, where (a)–(d) denote different representative test samples.

TABLE IV
CLASSIFICATION PERFORMANCE OF VARIOUS METHODS ON THE DDHRNET XI'AN DATASET UNDER CLOUD-COVERED CONDITIONS

Methods	Buildings	Roads	Greenery	Water	Farmland	mIoU	OA	Kappa
FTransUnet [65]	54.16	0.00	62.21	20.66	0.00	17.03	44.70	18.44
ASANet [28]	89.35	0.0	86.83	27.13	0.0	29.46	66.43	49.94
FuseNet [30]	80.76	33.52	79.04	68.45	43.96	48.99	72.83	61.71
CMFNet [66]	77.43	35.21	84.93	69.52	79.07	53.93	75.78	66.07
SA-Gate [26]	87.61	38.49	93.14	79.93	57.71	61.69	83.45	76.45
DCIFNet [67]	90.49	48.61	91.13	79.57	87.74	69.61	85.51	79.53
MFNet [64]	89.51	62.68	95.10	82.78	91.29	76.05	88.35	83.57
AdaptCloud	92.86	72.2	94.49	86.91	95.1	**80.57**	**90.95**	**87.29**

[see Fig. 10(c) and (d)], where information loss is severe, AdaptCloud leverages foundation-model priors together with token-category association and multimodal feature optimization to aggregate residual information and couple cross-modal features, significantly alleviating cloud-induced information loss. In contrast, FuseNet fails to exploit residual cues and produces increasing noise as cloud cover thickens, FTransUNet suffers from convergence difficulties due to the scarcity of cloud-covered samples and thus generates frequent misclassifications, while SA-Gate, although comparatively stronger, still shows road discontinuities and confusion between farmland and greenfield. DCIFNet degrades more sharply than AdaptCloud for confusable classes [e.g., roads versus buildings in Fig. 10(a)] as cloud occlusion intensifies, mainly because it captures discriminative cues less effectively under cloud-induced uncertainty. Fig. 10(d) further demonstrates AdaptCloud's sustained advantage under dense cloud conditions.

2) Results on DDHRNet Xi'an Dataset: The quantitative comparison results on the DDHRNet-Xi'an dataset are shown in Table IV. AdaptCloud achieves the highest accuracy, with mIoU improvements of 63.54%, 51.11%, 31.58%, 26.64%, 10.96%, 4.52%, and 3.51% in mIoU over FTransUNet, ASANet, FuseNet, CMFNet, SA-Gate, DCIFNet, and MFNet, respectively. These results highlight distinct differences in the utilization of information under cloud occlusion by each method. As shown in Table IV, the classification performance of FTransUNet, FuseNet, and CMFNet significantly

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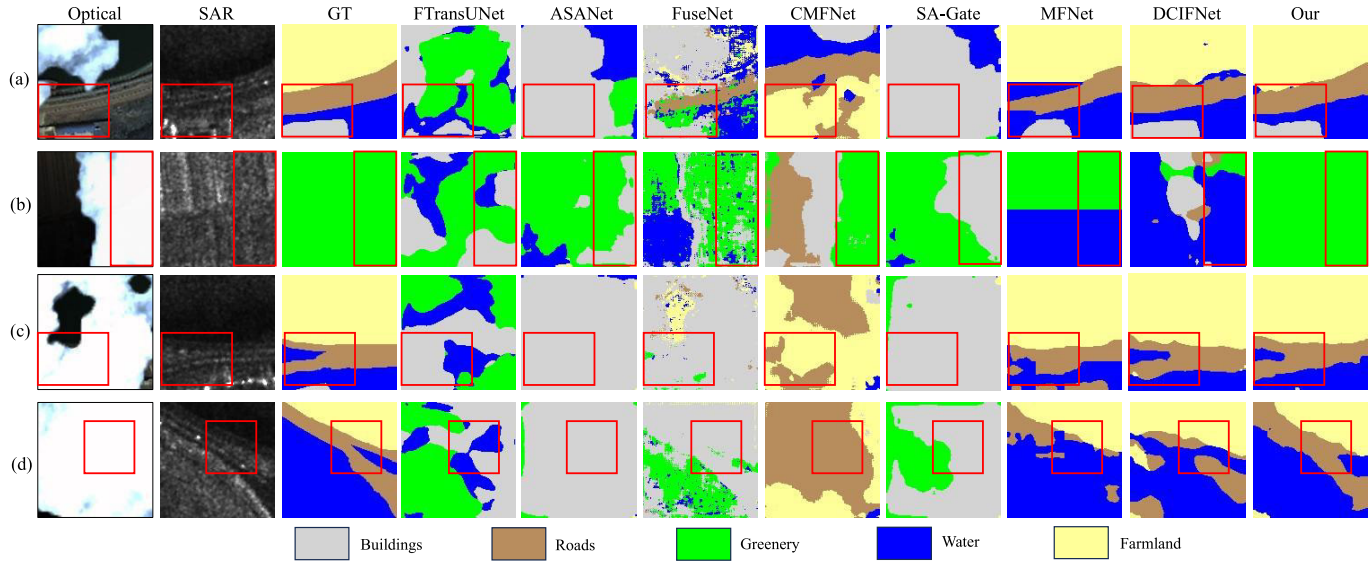


Fig. 11. Visualized classification results of DDHRNet Xi'an, where (a)–(d) denote different representative test samples.

TABLE V
CLASSIFICATION PERFORMANCE OF VARIOUS METHODS ON THE PIE-RGB-SAR DATASET UNDER CLOUD-COVERED CONDITIONS

Methods	Backbroad	City	Road	Water	Forest	Farmland	mIoU	OA	Kappa
FTransUNet [65]	32.48	63.99	0.00	50.10	63.60	14.13	28.42	53.35	32.12
MFNet [64]	96.83	1.42	0.49	30.68	86.63	3.40	22.32	47.17	22.37
CMFNet [66]	95.58	88.56	0.01	57.97	91.14	0.21	55.76	76.47	64.84
FuseNet [30]	97.85	86.43	19.28	55.49	87.63	18.34	52.34	76.77	65.71
SA-Gate [26]	97.03	87.72	22.01	57.64	58.85	85.31	53.45	71.76	62.45
DCIFNet [67]	98.03	90.02	29.59	70.89	86.31	49.05	61.30	81.73	73.88
ASANet [28]	97.92	90.16	58.05	78.03	90.13	73.75	72.38	87.25	81.94
AdaptCloud	98.29	90.85	66.96	82.34	90.69	80.75	**76.25**	**89.00**	**84.50**

deteriorates under cloud occlusion, indicating that these methods fail to fully utilize residual information in the cloud-covered regions, leading to a sharp decline in performance. MFNet and SA-Gate demonstrate enhanced adaptability by integrating prior knowledge from pretrained models; yet, they remain insufficient in leveraging cloud degradation information resulting in accuracy declines of at least 4.52% on the mIoU metric compared with AdaptCloud. DCIFNet substantially improves the building classification under cloud occlusion through structural-consistency interaction and achieves 90.49% accuracy; yet, its road recognition remains limited in confusable scenarios, such as roads versus buildings, indicating that its differential structural-consistency guidance underutilizes stable low-frequency optical cues under clouds and therefore does not fully exploit discriminative information. Under cloud-obscured conditions, AdaptCloud significantly improved road classification accuracy by at least 9.52% on mIoU, effectively validating the advantage of SIAC in incorporating SAR texture details. Due to cloud cover, farmland often suffers from increased noise interference, leading to reduced accuracy. However, as shown by the classification results, our method achieves at least a 3.81% improvement in mIoU, indicating that ISFO effectively extracts useful features and enhances the stability of reliable information under cloud cover.

The visualization results of each model on the DDHR-Net Xi'an dataset are shown in Fig. 11, where the image sequence includes RGB images, SAR images, ground truth labels, and the results from FTransUNet, CMFNet, ASANet, FuseNet, MFNet, DCIFNet, and AdaptCloud. These results demonstrate the superiority of AdaptCloud, particularly in road recognition, where it shows better continuity. As shown in Fig. 11, FTransUNet is sensitive to SAR images, leading to confusion between Greenery and farmland. CMFNet significantly deteriorates under thick cloud occlusion, revealing poor resistance to cloud interference. ASANet and DCIFNet use differentiated interaction for deep fusion to alleviate cloud occlusion; yet, they still confuse water and farmland, with errors increasing in the more occluded areas of Fig. 11(c) and (d), indicating a lack of targeted optimization for confusable classes under clouds. AdaptCloud fully integrates optical and SAR stability information under cloud cover conditions and enhances interclass separability by combining class association fine-tuning strategies, thus demonstrating strong resistance to cloud interference.

3) Results on PIE-RGB-SAR Dataset: The quantitative comparison results between AdaptCloud and other methods on the PIE-RGB-SAR dataset are presented in Table V. AdaptCloud achieves the best classification performance, with mIoU scores surpassing MFNet, FuseNet, DCIFNet, SA-Gate,

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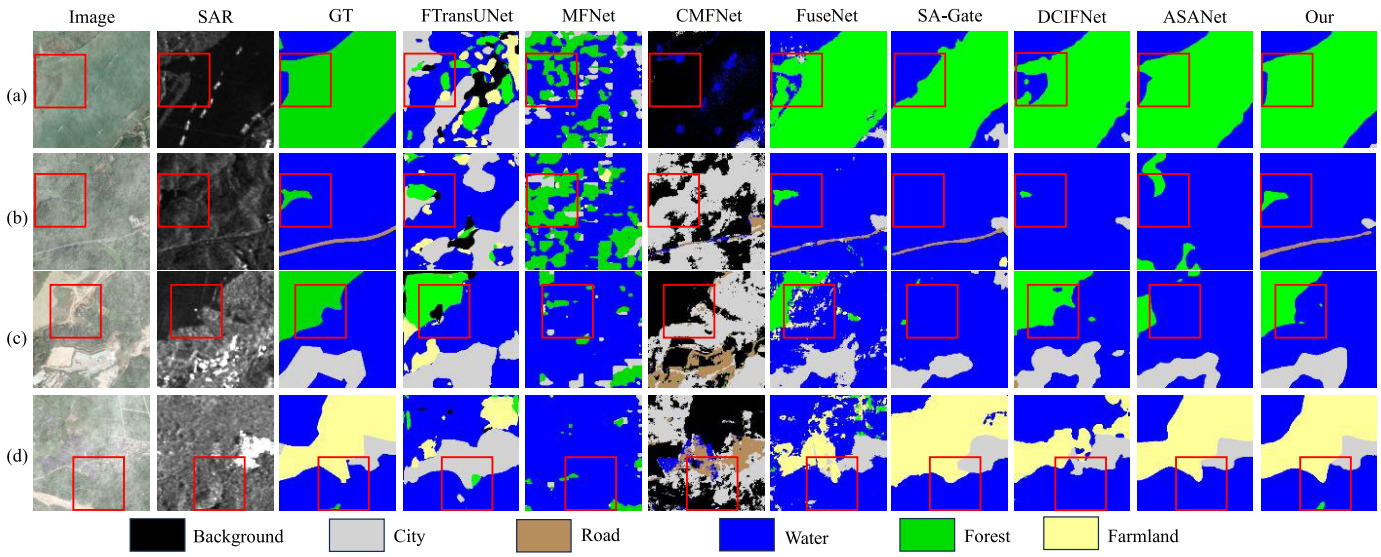


Fig. 12. Visualized classification results of PIE-RGB-SAR, where (a)–(d) denote different representative test samples.

TABLE VI
CLASSIFICATION PERFORMANCE OF VARIOUS METHODS ON THE CSU-OPT-SAR DATASET UNDER CLOUD-COVERED CONDITIONS

Methods	Farmland	Greenery	Impervious	Road	Water	mIoU	OA	Kappa
FTransUNet [65]	30.54	74.24	39.32	2.36	24.16	21.90	48.26	21.98
CMFNet [66]	52.64	82.80	58.41	1.79	24.42	30.77	60.08	40.81
FuseNet [30]	44.52	87.35	63.09	28.09	39.69	39.73	64.98	47.75
SA-Gate [26]	48.38	86.69	55.36	48.13	47.52	42.27	66.60	51.63
MFNet [64]	69.16	87.27	71.65	45.17	52.30	51.85	73.98	62.54
DCIFNet [67]	78.96	87.12	79.64	57.92	69.08	60.99	80.03	71.91
ASANet [28]	80.36	88.92	78.76	57.05	67.97	62.21	80.78	72.74
AdaptCloud	79.82	89.54	78.79	59.48	68.48	**62.92**	**81.20**	**73.31**

and ASANet by 42.42%, 23.91%, 14.95%, 13.31%, and 3.87% in mIoU, respectively. The low performance of FuseNet, with only 52.34% mIoU, indicates its weak resistance to cloud interference. In addition, the scarcity of road samples and their susceptibility to cloud occlusion make classification difficult, especially in areas with high cloud coverage in this dataset. However, AdaptCloud still achieves strong results, demonstrating the advantage of using underlying prior knowledge and category-specific trainable tokens to effectively focus on residual information under cloud occlusion. Due to the spectral similarity between forests and farmlands, they are prone to confusion, particularly under cloud occlusion, and obtaining samples is challenging. Nonetheless, AdaptCloud method achieves the highest classification accuracy for these two classes, demonstrating the effective use of multimodal stable priori knowledge by ISFO and the efficiency of SIAC in multimodal feature coupling.

Fig. 12 presents the classification visualization results on the PIE-RGB-SAR dataset, where (a)–(d) correspond to images with progressively increasing cloud cover. As shown in Fig. 12, AdaptCloud demonstrates stable and superior performance under various cloud cover conditions, indicating that the proposed method can effectively extract usable information from cloud-obscured areas. In Fig. 12(b), FuseNet, FTransUNet, and MFNet exhibit significant confusion between farmland and forest, despite their distinct textural characteristics. Cloud cover severely degrades optical textures, and these methods fail to fully leverage stable information beneath cloud resulting in a significant decline in classification accuracy under cloudy conditions. In Fig. 12(d), although DCIFNet utilizes differential priors to guide optical feature extraction, its classification performance under cloud cover is limited due to the SAR branch's lack of comprehensive spectral low-frequency information, such as tonal variations, for water bodies and croplands. AdaptCloud maintains relatively clear road continuity in such scenarios, owing to SIAC's capability to recover missing observations and ISFO's implementation of stable cross-modal feature fusion.

4) Results on CSU-OPT-SAR Dataset: Table VI summarizes the quantitative comparison results on the CSU-OPT-SAR dataset. The best classification performance was achieved by AdaptCloud, with its mIoU values improving by 41.02%, 32.15%, 23.19%, 20.65%, 11.07%, 1.93%, and 0.71% over FTransUNet, CMFNet, FuseNet, SA-Gate, MFNet, DCIFNet, and ASANet, respectively. FuseNet shows the subpar performance, with its mIoU 42.24% lower than AdaptCloud's which reflects limited capability in utilizing residual information and weaker resistance to cloud interference. ASANet demonstrates strong performance, but under cloud cover, its failure to fully utilize SAR texture information results in a 2.43% lower road classification accuracy compared with our method. DCIFNet acquires high-frequency structural

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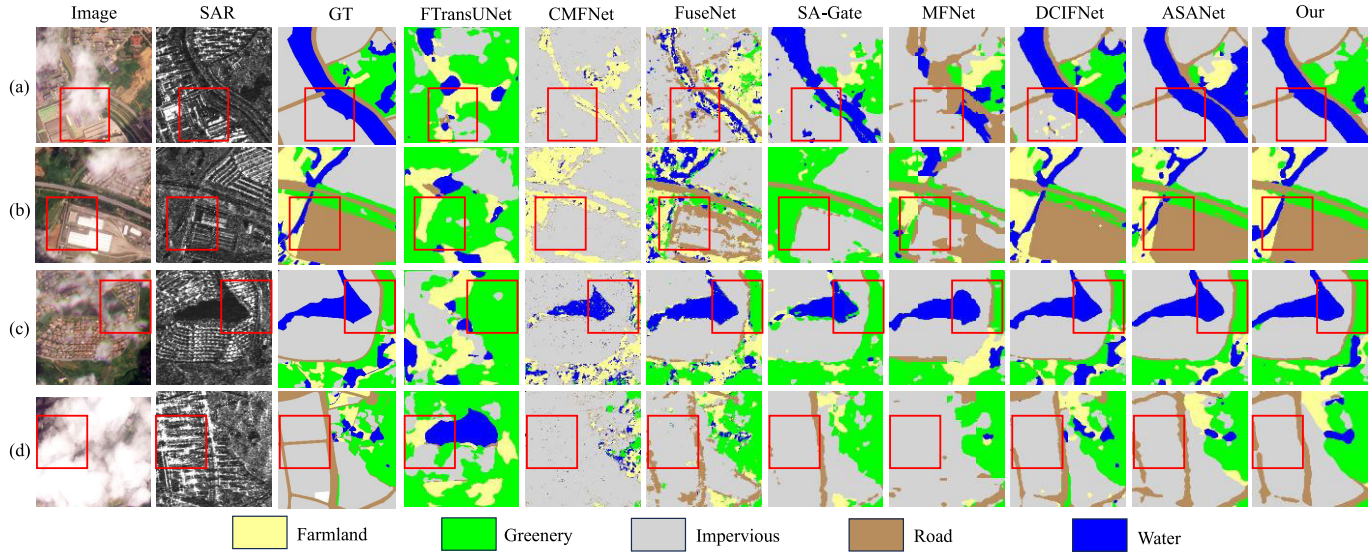


Fig. 13. Visualized classification results of CSU-OPT-SAR, where (a)–(d) denote different representative test samples.

consistency information through bidirectional differencing, enabling SAR high-frequency structures to guide optical classification with overall satisfactory results. However, the low-frequency structural information in optical imagery is not fully leveraged to guide SAR feature extraction, resulting in classification performance for certain easily confused objects (such as farmland and greenery) that falls short of our method. These results confirm that AdaptCloud utilizes limited information under cloud cover to achieve stronger and more reliable LCC performance.

Fig. 13 illustrates the classification results on the CSU-OPT-SAR dataset, with (a)–(d) being images under progressively increasing levels of cloud cover. The results show that AdaptCloud consistently outperforms competing methods across different cloud conditions, highlighting its superior capacity to exploit limited information. FTransUNet and CMFNet exhibit pronounced confusion between roads and buildings, a consequence of their reliance on degraded optical data and insufficient use of SAR information for effective multimodal coupling. As shown in Fig. 13(a)–(d), FuseNet, SA-Gate, and MFNet face challenges in distinguishing spectrally similar features, such as roads and buildings, thereby emphasizing the importance of incorporating SAR texture details. DCFNet and ASANet demonstrate good overall classification performance but show limitations in road connectivity and misclassify some buildings as farmland, highlighting the importance of focusing on stable key features under cloud occlusion conditions. In contrast, AdaptCloud achieves the accurate differentiation in these cases, demonstrating its ability to selectively exploit reliable SAR features through ISFO while maintaining robust classification under cloud occlusion.

D. Land Cover Mapping Results Assessment

To comprehensively evaluate AdaptCloud's cloud-robust land cover mapping capabilities, we conducted both in-domain and cross-domain mapping analyses. Fig. 14 presents the LCC results in-domain. As shown in Fig. 14(c) and (f), the road skeleton remains connected and clearly delineated even in areas with thin to locally thick clouds. This demonstrates that AdaptCloud significantly enhances the extraction and utilization of limited observational information under cloud-obscured conditions by reinforcing optical low-frequency structures and SAR high-frequency textures, while selectively suppressing cloud-induced instability responses through cross-modal fusion. In addition, as shown in Fig. 14(c), within cropland and green space transition zones, the model yields more stable detection of scattered buildings, with markedly fewer misses and misclassifications. In winter scenes, despite the strong spectral similarity between cropland and grassland [see Fig. 14(d)], the model leverages robust, cross-modally consistent cues to suppress the spread of uncertain regions, thereby improving boundary consistency and class separability. Furthermore, to evaluate the cross-domain mapping capability of the AdaptCloud method under cloud cover, we conducted mapping analysis on the CSU-OPT-SAR-Unlabeled dataset. Despite significant differences in cloud cover and data style between Figs. 14 and 15, the experimental results in Fig. 15 demonstrate that narrow roads and impervious surfaces are clearly identified, confirming the strong generalization capability of AdaptCloud. Fig. 15(c) demonstrates that despite the challenges of distinguishing scattered farmland from green areas in hilly regions, AdaptCloud accurately classifies these complex land cover types, showcasing its ability to effectively utilize limited data in cloud-obscured environments. Overall, AdaptCloud not only performs exceptionally well in the region of the training dataset but also demonstrates remarkable generalization capabilities in cross-domain transfer tasks, offering strong support for high-accuracy land cover mapping in cloud-obscured environments.

E. Ablation Study

To analyze each key module's contribution to cloud-covered LCC within the AdaptCloud method, an ablation experiment was conducted. The PIE-RGB-SAR dataset, with its diverse

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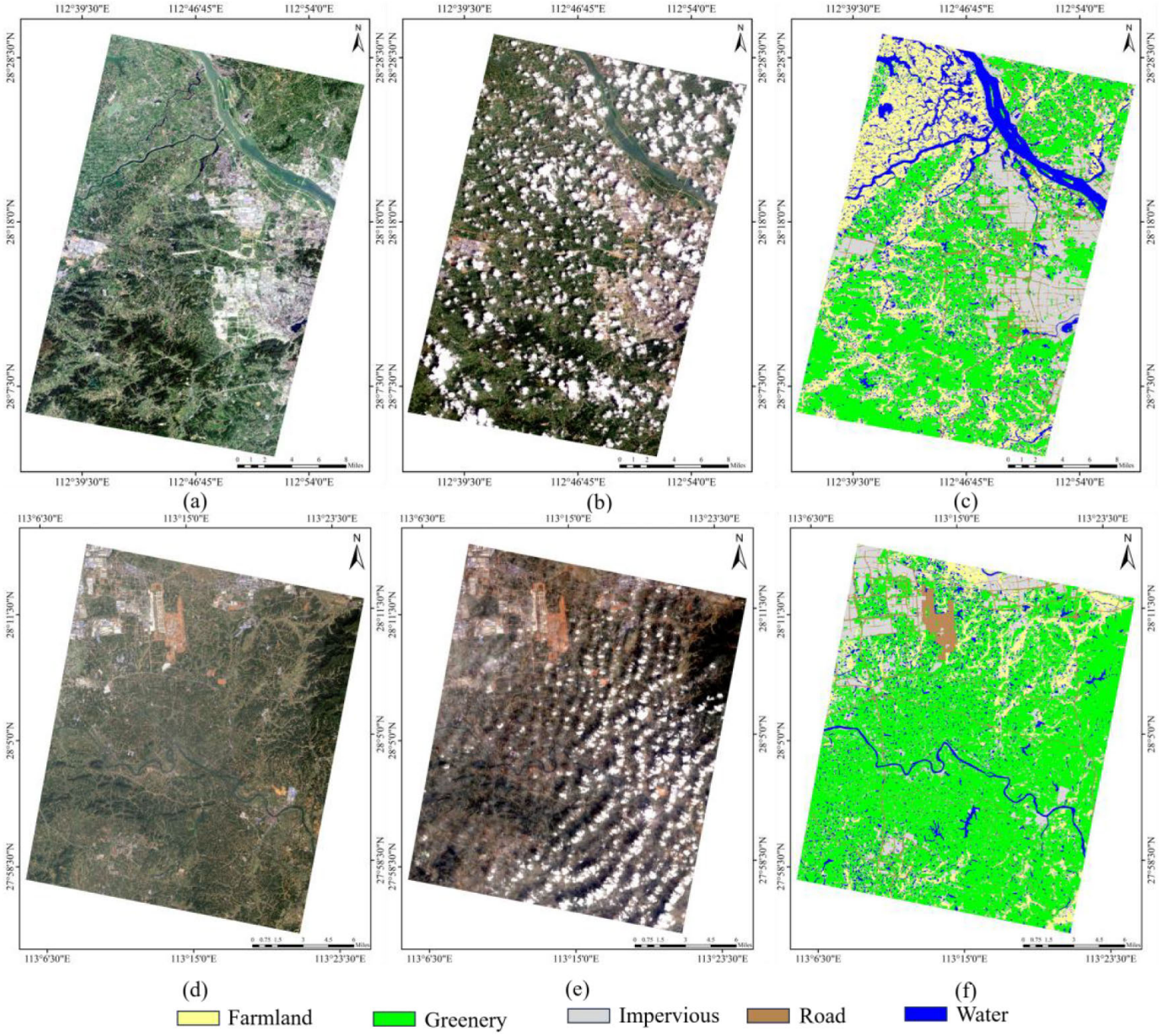


Fig. 14. Land cover mapping results based on CSU-OPT-SAR labeled data under cloud cover. (a) CSU-OPT-SAR summer no cloud optical image. (b) CSU-OPT-SAR summer optical image under cloud cover. (c) Land cover mapping results of CSU-OPT-SAR summer under cloudy cover. (d) CSU-OPT-SAR winter no cloud optical image. (e) CSU-OPT-SAR winter optical image under cloud cover. (f) land cover mapping results of CSU-OPT-SAR winter under cloud cover.

TABLE VII
ABLATION EXPERIMENT ON PIE-RGB-SAR AND DDHRNET XI'AN

Modality	FICDS	ISFO	SIAC	PIE-RGB-SAR			DDHRNet Xi'an		
				mIoU (%)	OA (%)	Kappa (%)	mIoU (%)	OA (%)	Kappa (%)
SAR	---	---	---	63.33	82.04	74.52	41.98	56.12	41.87
RGB	---	---	---	68.65	85.34	79.38	67.90	84.82	78.78
RGB+SAR	✓			72.92	87.55	82.46	72.86	86.53	81.28
RGB+SAR	✓	✓		73.14	87.62	82.57	74.83	87.62	82.71
RGB+SAR	✓	✓	✓	**75.19**	**88.67**	**84.08**	**80.57**	**90.95**	**87.29**

cloud cover types, and the DDHRNet Xi'an dataset, featuring dense cloud coverage, provide a comprehensive testbed for assessing performance under cloud-obscured conditions. Ablation experiments on these datasets validate the effectiveness of our method, showcasing significant improvements over single-modality inputs. For instance, as shown in Table VII, the

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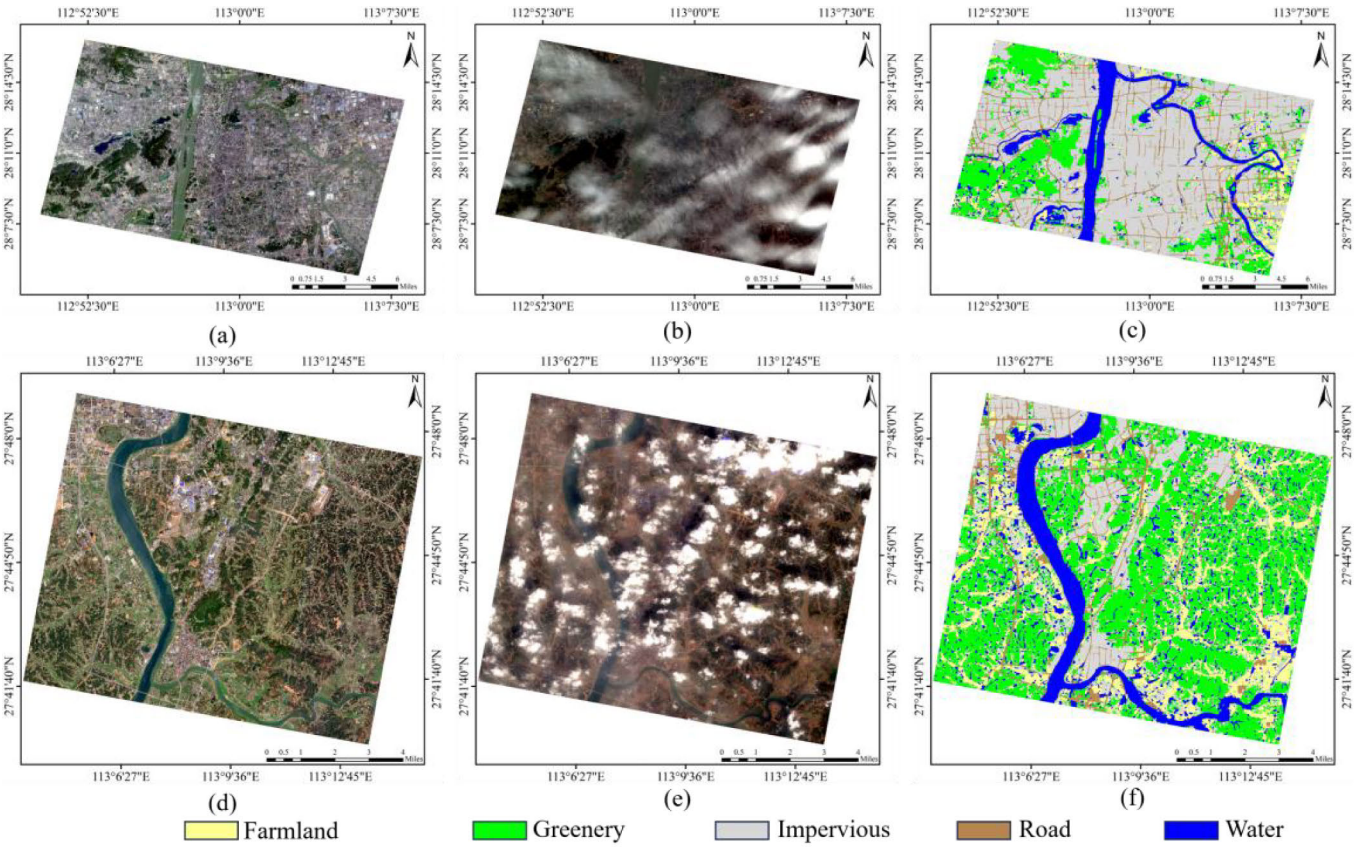


Fig. 15. Land cover mapping results based on CSU-OPT-SAR-Unlabeled data under cloud cover. (a) CSU-OPT-SAR-Unlabeled Changsha no cloud optical image. (b) CSU-OPT-SAR-Unlabeled Changsha optical image under cloud cover. (c) Land cover mapping results of CSU-OPT-SAR-Unlabeled Changsha under cloud cover. (d) CSU-OPT-SAR-Unlabeled Zhuzhou no cloud optical image. (e) CSU-OPT-SAR-Unlabeled Zhuzhou optical image under cloud cover. (f) Land cover mapping results of CSU-OPT-SAR-Unlabeled Zhuzhou under cloud cover.

mIoU increased by over 4.27%, demonstrating that our framework outperforms both optical and SA single-modality inputs. FICD enhances the multimodal feature extraction by leveraging prior knowledge from the foundation model, especially in cloudy scenarios. On both PIE-RGB-SAR and DDHRNet Xi'an datasets, the SIAC module improved accuracy, with a notable increase in mIoU by 1.97% on DDHRNet Xi'an. The ISFO module brought even greater improvements, especially on the DDHRNet Xi'an dataset, where it increased mIoU by 5.74%, OA by 3.35%, and Kappa by 4.58%. This demonstrates how SIAC optimizes the use of low-frequency structure optical and high-frequency texture SAR features, reinforcing multimodal coupling and improving overall classification accuracy.

V. Discussion

To provide a comprehensive evaluation of AdaptCloud, we analyze its performance from four perspectives. First, to assess its adaptability to cloudy conditions, we compare AdaptCloud with existing SOTA methods under both cloud-contaminated and cloud-free scenarios. We further investigate the effects of the token number and the Top-K sparsity ratio in ISFO on model performance. Second, to evaluate the model size and computational complexity, we compare the number of parameters and FLOPs of AdaptCloud with those of all baseline methods. Finally, to assess its temporal consistency and real-world applicability, we systematically compared the generated LCC results with the existing 2023 CLDC (30-m resolution) [69] and SinoLC-1 (1-m resolution) [11] products.

A. Impact of Cloud Occlusion on Method Performance

To evaluate the performance degradation of different methods under cloud cover conditions, this article systematically compares the classification accuracy of four datasets in both cloud and no cloud scene. As shown in Fig. 16, the experimental results indicate that cloud occlusion significantly impacts model performance, leading to a general decline in classification accuracy. Specifically, FTransUnet, CMFNet, MFNet, and FuseNet, while employing multimodal feature fusion, primarily rely on global features and simple fusion mechanisms between modalities. These methods are not specifically designed to address the information degradation caused by cloud cover, leading to performance degradation under such conditions, with mIoU dropping by up to 13.3%. ASANet utilizes an asymmetric feature fusion strategy to exploit complementary features between optical and SAR images, but it fails to adequately handle the dynamic variations caused by cloud occlusion, resulting in performance fluctuations across different cloud conditions, with mIoU degradation ranging from 4% to 12.8%. DCIFNet mitigates cloud occlusion interference through structural consistency

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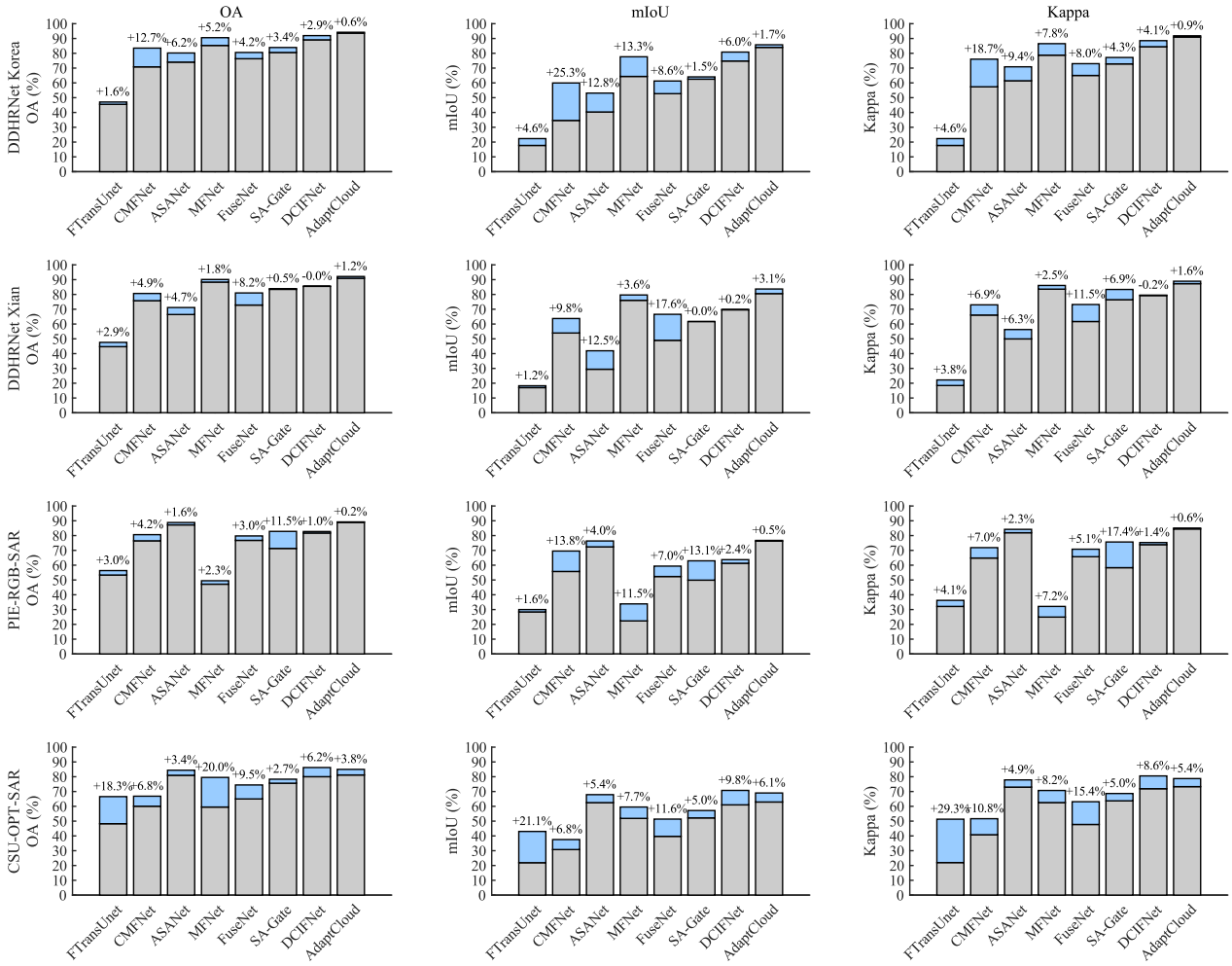


Fig. 16. Classification performance under cloud cover and cloud-free conditions.

and SAR guidance, demonstrating a degree of robustness. However, due to its incomplete modeling of the information instability caused by cloud occlusion, its OA remains inferior to AdaptCloud. SA-Gate employs an adaptive gating mechanism for multimodal fusion, yet in complex cloud and land cover environments, it fails to effectively leverage stable information under cloud occlusion, leading to poorer performance on the CSU-OPT-SAR dataset, with mIoU dropping 20.65% compared with AdaptCloud. In contrast, AdaptCloud, through an information-driven adaptive multimodal feature fusion approach, effectively utilizes the limited information under cloud occlusion, maintaining stable performance with a maximum mIoU decrease of only 6.1% and a minimum decrease of 0.5%, achieving optimal classification results in both cloud and no cloud scene, thus demonstrating its stability, robustness, and exceptional adaptability under cloud cover conditions.

B. Analysis of Token Number on Method Performance

To evaluate the impact of the number of tokens and the Top-K parameter k in AdaptCloud on LCC performance under cloud cover conditions, we conducted a parametric sensitivity analysis of the number of tokens. Given that the DDHRNet Korea dataset covers a wide range of thin and thick cloud scenarios, we chose this dataset for our experimental verification. As shown in Fig. 17, the results are generally stable when the token number ranges from 10 to 80, with variations of no more than 0.14% in OA, 0.35% in mIoU, and 0.19% in Kappa, indicating low sensitivity to m. Although m = 80 achieves the highest mIoU (85.59%), the gain over m = 50 is only 0.09%

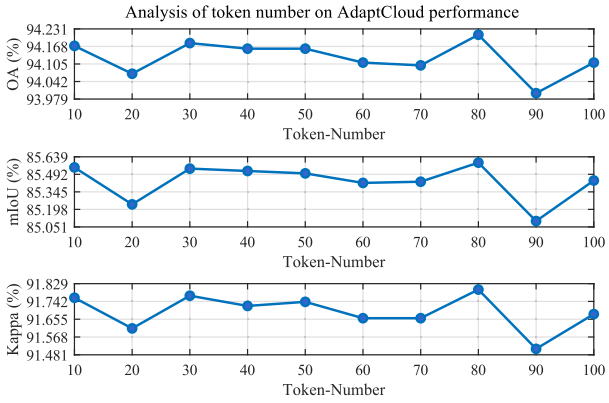


Fig. 17. Sensitivity analysis of token number on DDHRNet Korea.

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TABLE VIII
MODEL COMPUTATIONAL COST AND MODEL SIZE ANALYSIS ON THE DDHRNET KOREA DATASET

Method	FTransUNet	CMFNet	FuseNet	SA-Gate	ASANet	MFNet	DCIFNet	Our
FLOPs (G)	30.51	80	65.21	23.93	1.97	15.862	40.23	97.46
Parameters (M)	203.40	104.07	55.17	130.69	65.40	8.959	198.43	74.24

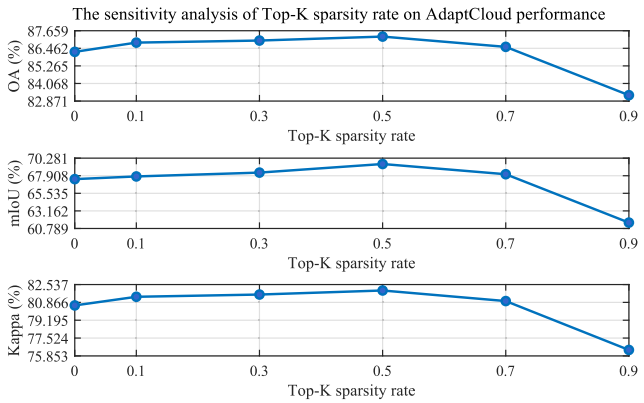


Fig. 18. Sensitivity analysis of Top-K sparsity rate on DDHRNet Korea.

(with OA and Kappa improving by just 0.05% and 0.06%), which is marginal. Considering the additional computational and parameter cost as m increases, and the performance drop observed at larger m (e.g., 90), we set m = 50 as the default to achieve a more robust tradeoff between accuracy and efficiency and to avoid dataset-specific tuning.

C. Analysis of Top-K Sparsity Impact on Methods Performance

To assess the impact of the Top-K sparsity rate on model performance, a parameter sensitivity analysis was conducted using the DDHRNet Korea dataset, which contains diverse thick and thin cloud imagery to simulate complex occlusion scenarios. To isolate its effect on classification, all other core modules were removed, retaining only the Top-K sparse attention component. First, we evaluate performance without sparsity enhancement. Subsequently, we conduct a sensitivity analysis of performance starting from a sparsity rate of 0.1, incrementing in steps of 0.2 up to 0.9. As shown in Fig. 18, as the sparsity rate increases, the model performance first improves and then declines. When the sparsity rate is within 0.0–0.5, OA, mIoU, and Kappa increase by 1.04%, 2.04%, and 1.4%, respectively, indicating that moderate sparsity effectively suppresses cloud-affected cross-modal correspondences while preserving stable optical–SAR feature interactions, thereby yielding a more robust multimodal representation. At a sparsity rate of 0.5, the model achieves optimal performance with an OA of 87.26%, mIoU of 69.49%, and Kappa of 81.98%, indicating that cloud occlusion still results in significant information loss. When the sparsity rate exceeds 0.5 (0.5–0.9 range), performance declines markedly, with OA, mIoU, and Kappa decreasing by 3.99%, 7.91%, and 5.57%, respectively, indicating that excessive sparsity discards valid cross-modal correspondences and undermines the integrity of multimodal feature fusion. Based on the above analysis, a sparsity intensity within 0–0.5 is optimal for mitigating cloud occlusion, ensuring stable enhancement of critical spectral features across modalities.

D. Analysis of Computational Cost and Model Size

To systematically evaluate the computational overhead of AdaptCloud, we report the parameter counts and GFLOPs of all compared methods. Given the space limit and the DDHRNet Korea dataset's extensive coverage of both thin and thick cloud conditions, this section presents a model complexity analysis on DDHRNet Korea under an identical experimental setting. As reported in Table VIII, ASANet requires only 1.97G FLOPs, which is the lowest among all compared methods, while MFNet maintains a low computational cost of 15.862G FLOPs with only 8.959 M parameters. By comparison, FTransUNet and DCIFNet exhibit significantly larger parameter scales, with 203.40 and 198.43 M parameters, respectively. Our method contains 74.24 M parameters, corresponding to approximately 0.365 times those of FTransUNet, 0.568 times those of SA-Gate and 0.713 times those of CMFNet. Compared with DCIFNet, our method uses approximately 0.374 times the number of parameters, corresponding to a 62.6% reduction, while requiring approximately 2.423 times the FLOPs. The results indicate that incorporating SIAC and ISFO within the foundation-model fine-tuning framework substantially increases the computational cost in terms of FLOPs relative to the compared methods. However, because ViT pretrained weights are adopted and the encoder is fully frozen, the overall parameter count remains largely unchanged.

E. Comparison of Cloud-Cover LCC With LCPs

To evaluate AdaptCloud's effectiveness in mitigating temporal inconsistencies in land-cover products, we used two national-scale benchmarks: the 30-m CLDC product from 2023 derived from multiseason imagery and the 1-m SinoLC-11 product assembled from multitemporal Google imagery. Both datasets closely match our temporal coverage or spatial resolution while achieving nationwide cloud-free mosaics via integration of historical scenes, thereby providing comparable baselines for temporal-inconsistency assessment. For fair comparison, we harmonized categories by mapping both products to the five-class taxonomy adopted in this article. As shown in Fig. 19, this area is mainly an urban area and is significantly affected by thin cloud cover, resulting in serious loss of image information, especially in the lower right corner, where the ground features are almost unrecognizable. Both our imagery and the CLDC national LCP are from 2023.

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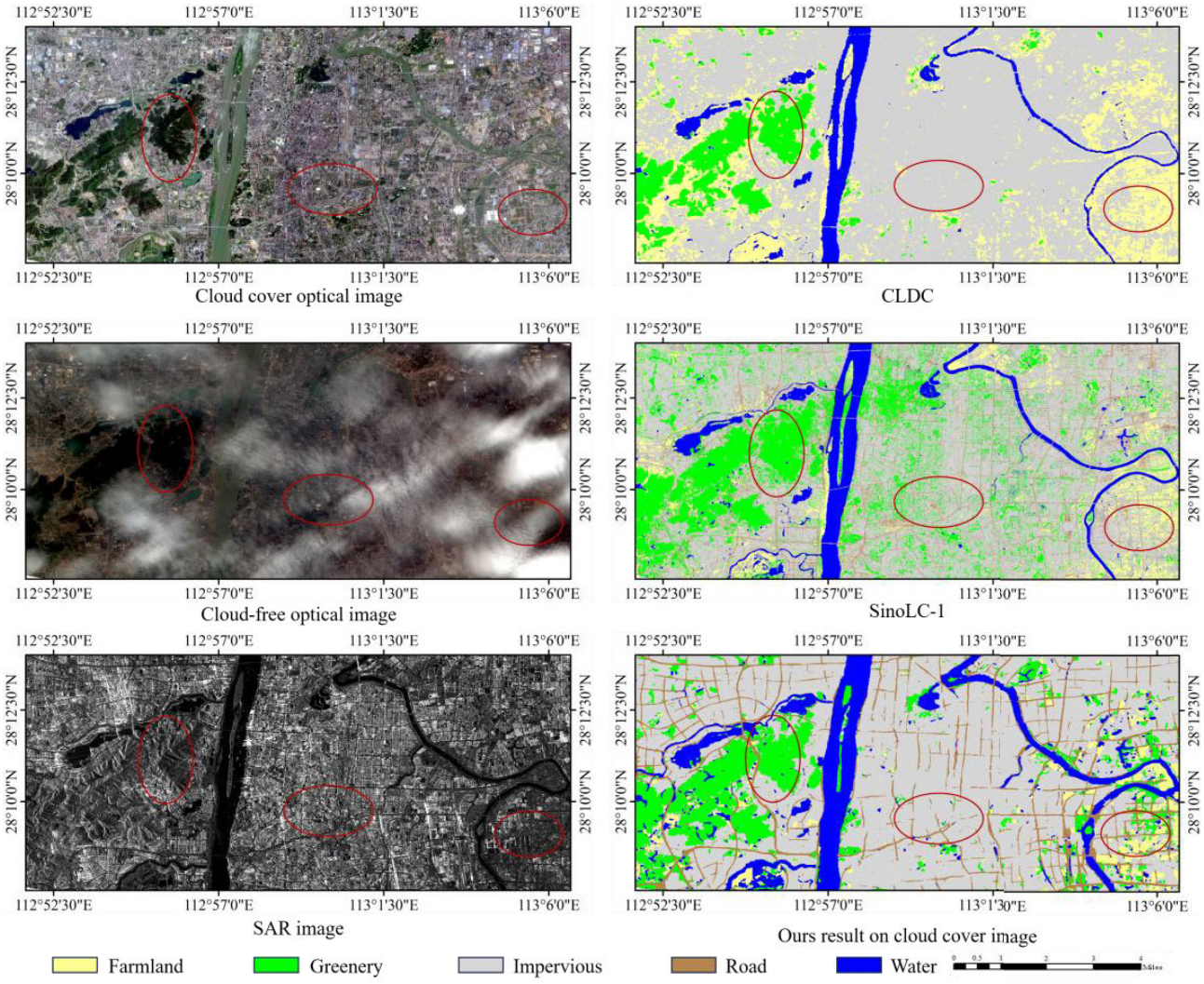


Fig. 19. Comparison with different land-use products.

However, analysis of the lower right corner reveals that this area is primarily impervious surface with a small amount of cultivated land; yet, the CLDC misclassifies it as having a large amount of cultivated land. Considering national farmland protection policies and the irrationality of converting impervious surface to cultivated land in a short period of time, the CLDC classification results exhibit significant bias. This demonstrates that while historical imagery is useful for large-scale land cover mapping, temporal inconsistencies can lead to errors in ground feature changes. Under cloud cover conditions, the AdaptCloud method can effectively distinguish impervious surfaces from other land features, fully demonstrating its superiority and application potential when data quality is poor. Despite the high spatial resolution (1 m) of the SinoLC-1 data product, significant misclassification of green vegetation occurred in the central-left region of the image, characterized by blurred boundaries and fragmented patches. However, the application of AdaptCloud markedly reduced these misclassification and fragmented patches issues. Even under cloud cover, AdaptCloud leverages the texture features of SAR imagery to enhance the accuracy of feature boundary detection.

VI. Conclusion

Cloud cover reduces LCC accuracy, and despite SAR–optical fusion enhancing penetration, occlusion still leads to incomplete information, increasing instability in multimodal modeling. This article reframes cloud cover as a limited-information condition and progressively improves category distinctiveness, observational integrity, and reliability. Specifically, we leverage foundation model prior information to enhance the feature extraction and integrate the category correlation optimization to improve class discrimination under cloud-obscured conditions. By integrating bidirectional information from optical structure and SAR texture prior cues, it enhances information integrity under cloud-obscured conditions and mitigates information loss caused by cloud cover. To address the issue of information instability caused by cloud cover uncertainty, we employ a cross-attention mechanism and sparse filtering method to enhance stable and critical features. In addition, to provide a robust benchmark for both cloud-free and cloud-contaminated scenarios the multiseasonal, multiregional, and multimodal CSU-OPT-SAR dataset was constructed. Experimental validation on DDHRNet Korea (thick and thin cloud covers), DDHRNet

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Xi'an (thick cloud cover), PIE-OPT-SAR (thick and thin cloud covers), and CSU-OPT-SAR (thick and thin cloud covers) achieved classification accuracies ranging from 81.20% to 93.60% using only 20% of the training samples, surpassing SOTA methods by 0.42%–8.29% in mIoU. AdaptCloud significantly enhances LCC under cloud cover, with strong potential for handling complex environments and informing future advancements in cloud-robust mapping methods.

However, in thick clouds and complex terrain, similar features like farmland and green spaces appear nearly identical in optical and SAR imagery. Their distinct geospatial distributions and category relationships, which are vital for accurate classification, are not effectively utilized by existing methods, resulting in unstable and unreliable outcomes. Future work will integrate more prior information to further model the effects of complex weather conditions.

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